



CMPT 413/713: Natural Language Processing

Constituency Parsing

Spring 2024
2024-03-04

Adapted from slides from Danqi Chen and Karthik Narasimhan
(with some content from Anoop Sarkar, David Bamman, Chris Manning,
Mike Collins, and Graham Neubig)

Overview

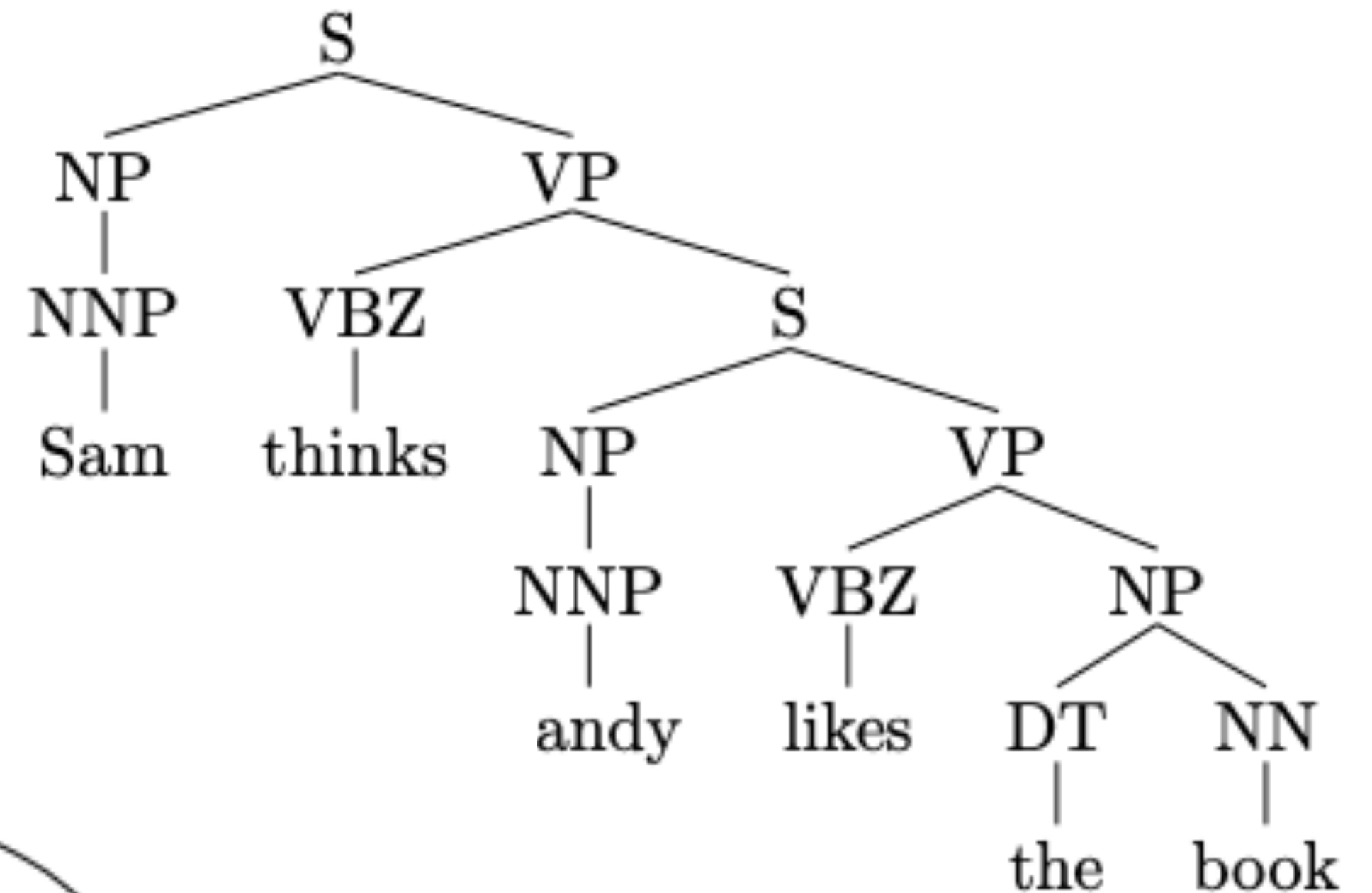
- Constituency structure vs dependency structure
- Context-free grammar (CFG)
- Probabilistic context-free grammar (PCFG)
- The CKY algorithm
- Evaluation
- Lexicalized PCFGs
- Neural methods for constituency parsing

Syntactic structure: constituency and dependency

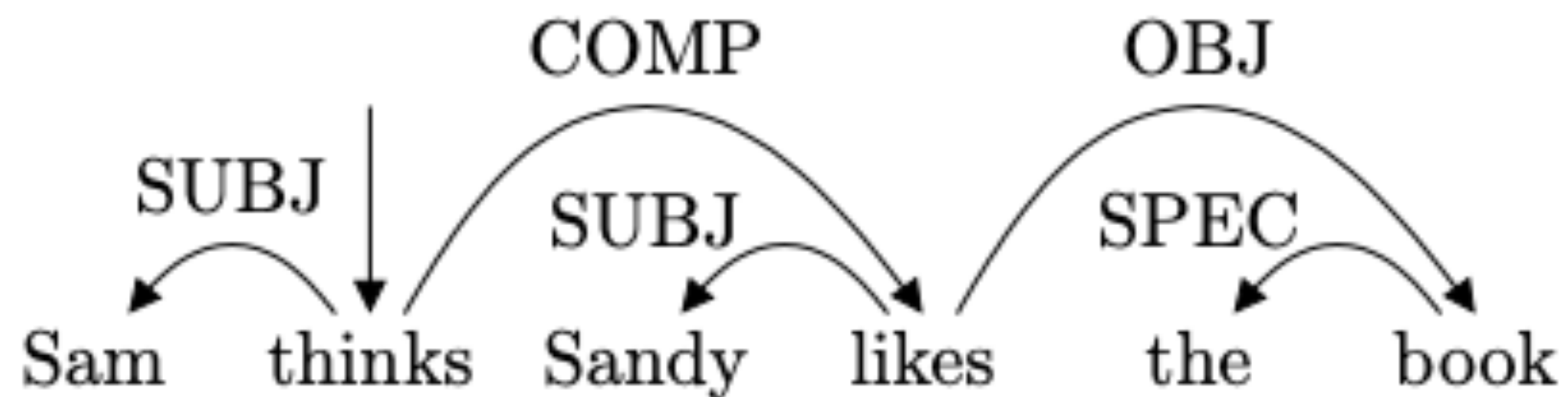
Two views of linguistic structure

- **Constituency**

- = phrase structure grammar
- = context-free grammars (CFGs)



- **Dependency**



Constituency structure

- **Phrase structure** organizes words into **nested constituents**
- Starting units: words **are given a category: part-of-speech tags**

the, cuddly, cat, by, the, door

DT, JJ, NN, IN, DT, NN

- Words combine into phrases **with categories**

the cuddly cat, by, the door

NP → DT JJ NN IN NP → DT NN

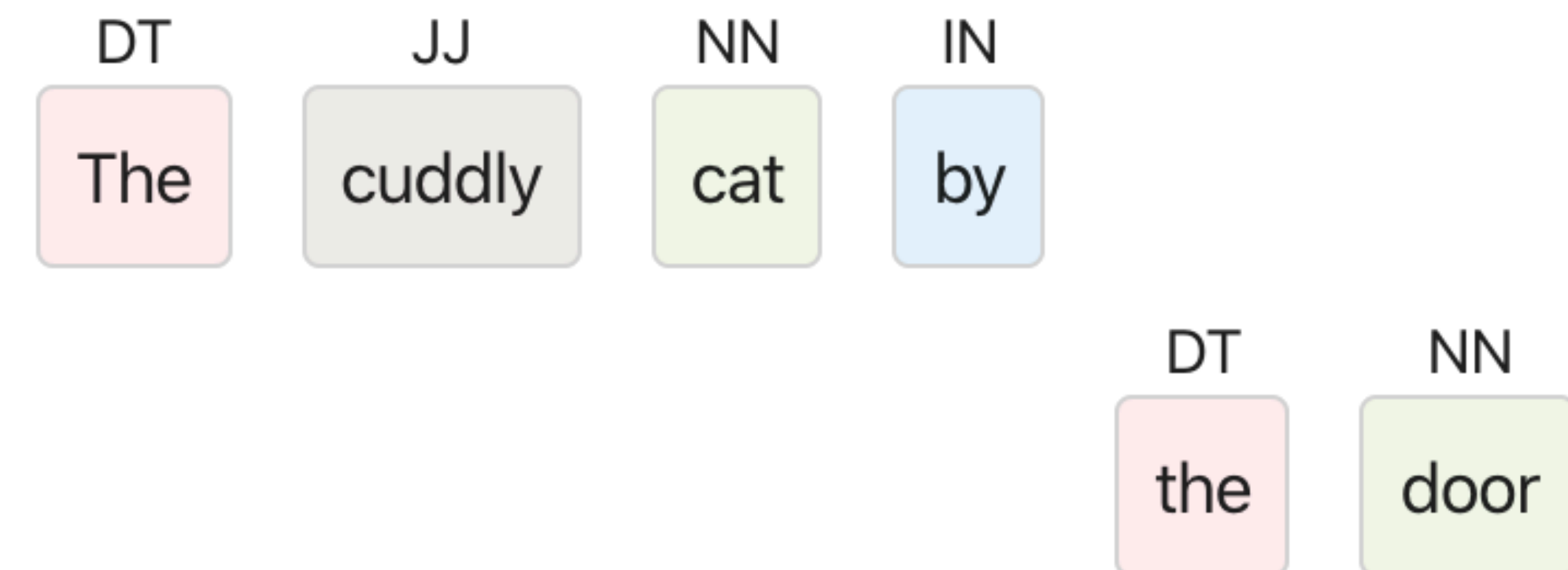
- Phrases can combine into bigger phrases **recursively**

the cuddly cat, by the door

NP PP → IN NP

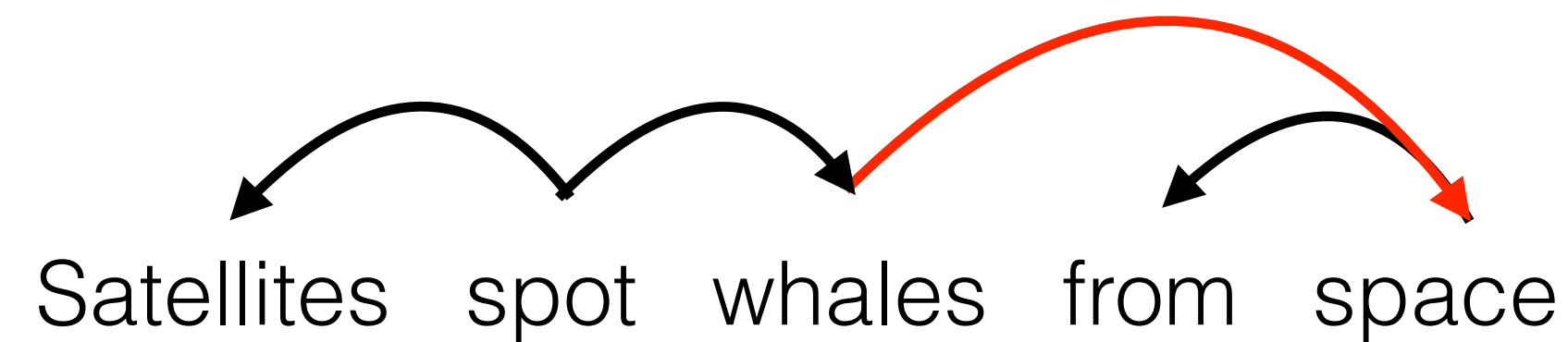
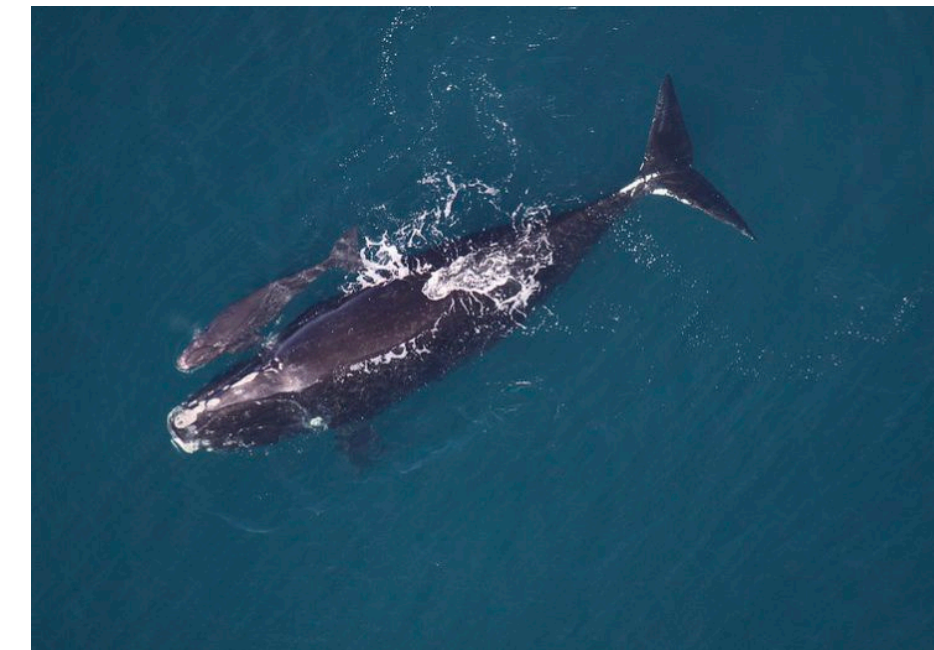
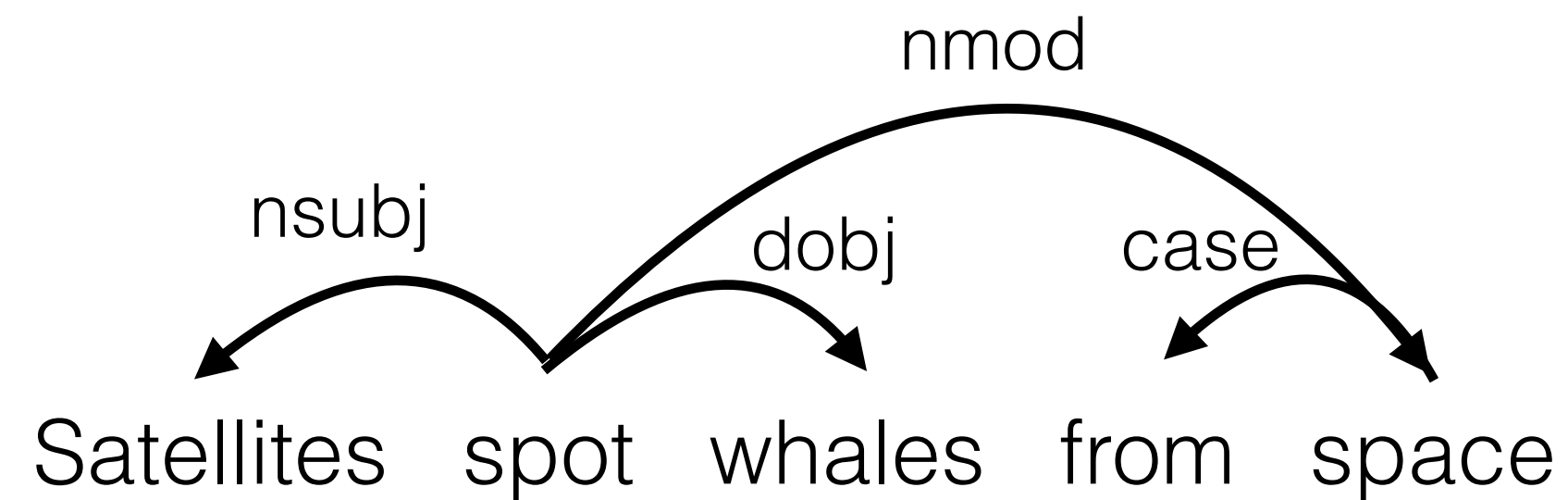
the cuddly cat by the door

NP → NP PP



Dependency structure

- Dependency structure shows which words depend on (modify or are arguments of) which other words.



Why do we need sentence structure?

- We need to understand sentence structure in order to be able to interpret language correctly
- Human communicate complex ideas by **composing** words together into bigger units
- We need to know what is connected to what

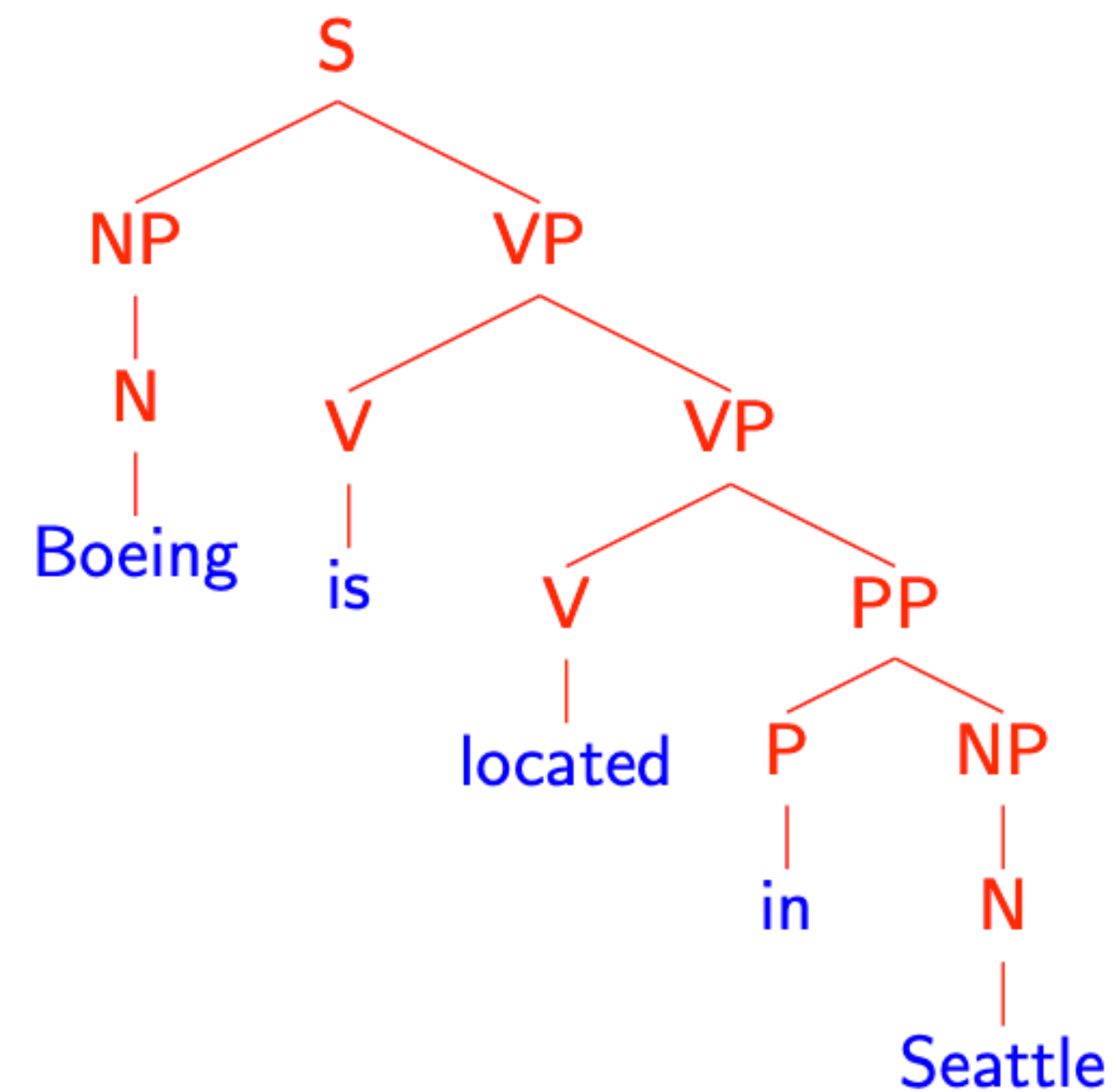
Syntactic parsing

- Syntactic parsing is the task of recognizing a sentence and assigning a **structure** to it.

Input:

Boeing is located in Seattle.

Output:

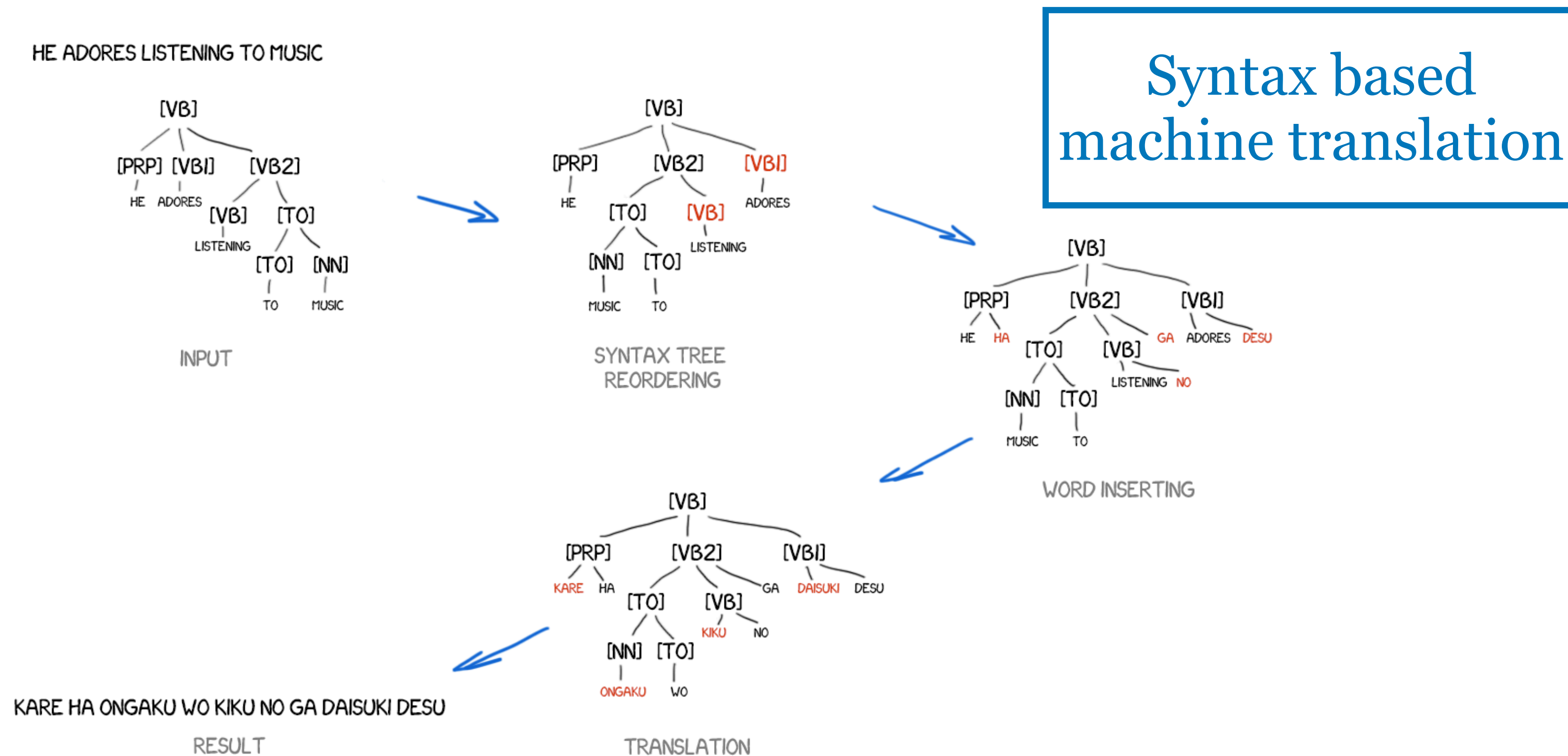


Syntactic parsing

- Used as intermediate representation for downstream applications

English word order: subject — verb — object

Japanese word order: subject — object — verb

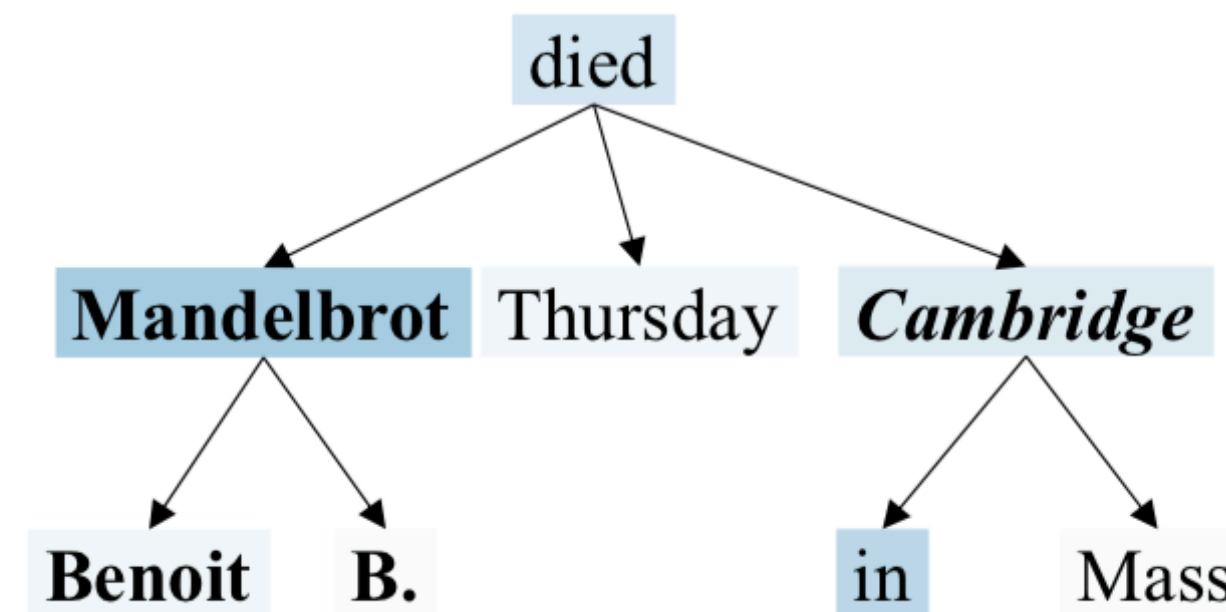


Syntactic parsing

- Used as intermediate representation for downstream applications

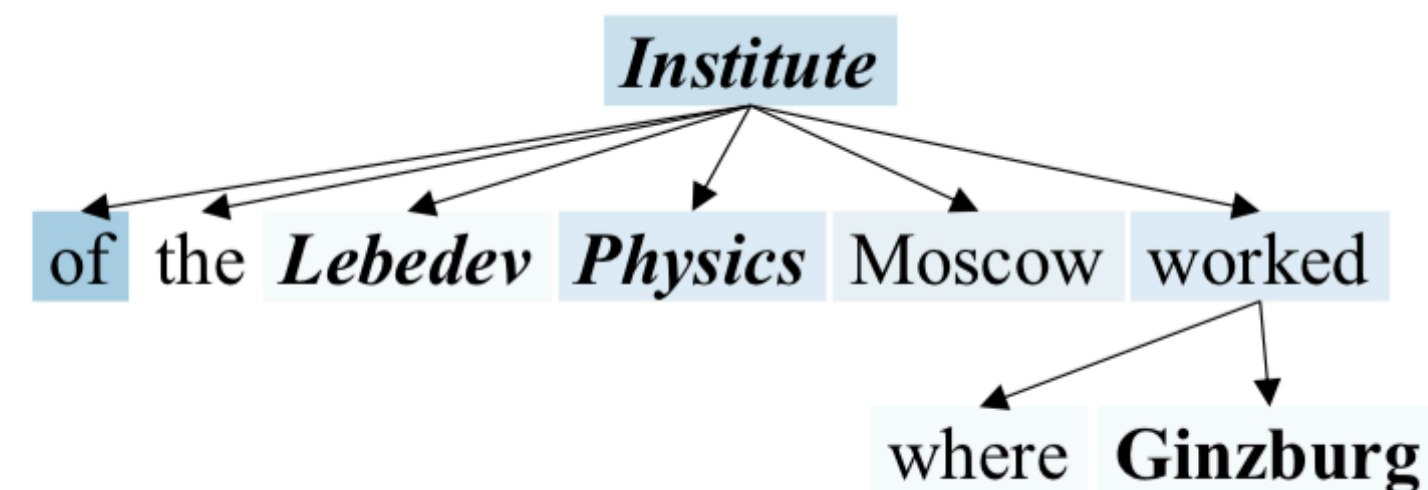
Relation: *per:city_of_death*

Benoit B. Mandelbrot, a maverick mathematician who developed an innovative theory of roughness and applied it to physics, biology, finance and many other fields, died Thursday in **Cambridge**, Mass.



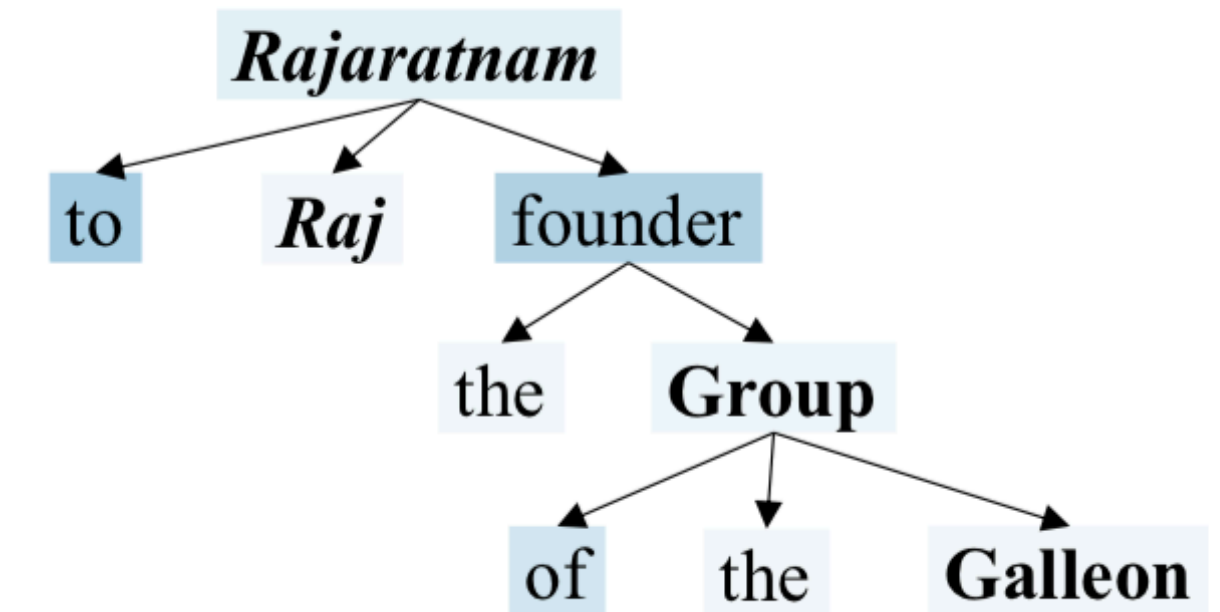
Relation: *per:employee_of*

In a career that spanned seven decades, Ginzburg authored several groundbreaking studies in various fields -- such as quantum theory, astrophysics, radio-astronomy and diffusion of cosmic radiation in the Earth's atmosphere -- that were of "Nobel Prize caliber," said Gennady Mesyats, the director of the **Lebedev Physics Institute** in Moscow, where **Ginzburg** worked .



Relation: *org:founded_by*

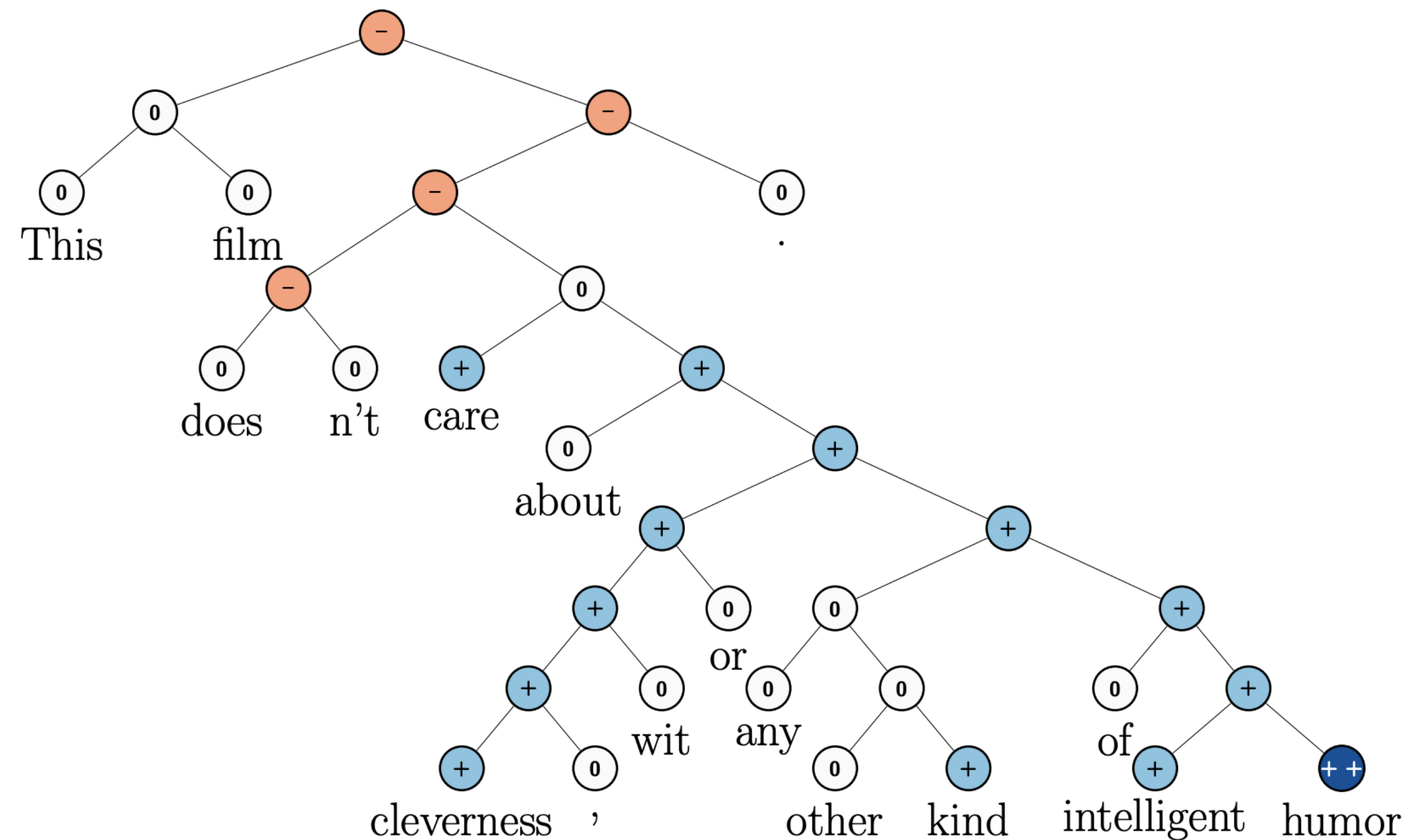
Anil Kumar, a former director at the consulting firm McKinsey & Co, pleaded guilty on Thursday to providing inside information to **Raj Rajaratnam**, the founder of the **Galleon Group**, in exchange for payments of at least \$ 175 million from 2004 through 2009.



Relation Extraction

Beyond syntactic parsing

This file doesn't care about cleverness, wit or any other kind of intelligent humor. **Negative**



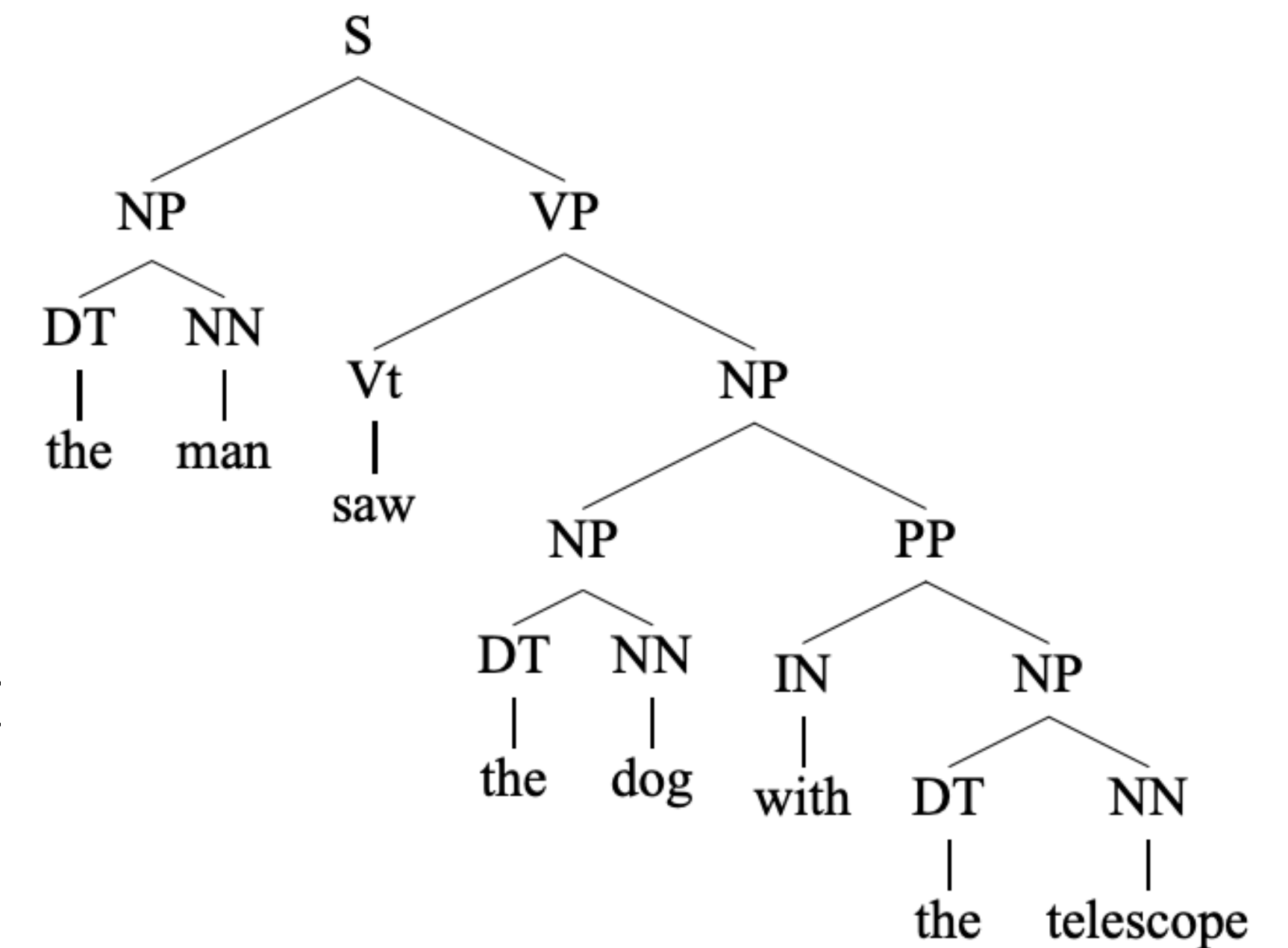
Nested Sentiment
Analysis

Recursive deep models for semantic compositionality over a sentiment treebank

Socher et al, EMNLP 2013

Context-free grammars (CFG)

- Widely used formal system for modeling constituency structure in English and other natural languages
- A context free grammar $G = (N, \Sigma, R, S)$ where
 - N is a set of **non-terminal** symbols
 - Σ is a set of **terminal** symbols
 - R is a set of **rules** of the form $X \rightarrow Y_1 Y_2 \dots Y_n$ for $n \geq 1, X \in N, Y_i \in (N \cup \Sigma)$
 - $S \in N$ is a distinguished **start** symbol



A Context-Free Grammar for English

$N = \{S, NP, VP, PP, DT, Vi, Vt, NN, IN\}$

$S = S$

$\Sigma = \{\text{sleeps, saw, man, woman, telescope, the, with, in}\}$
POS tags word

$R =$

S	→	NP	VP
VP	→	Vi	
VP	→	Vt	NP
VP	→	VP	PP
NP	→	DT	NN
NP	→	NP	PP
PP	→	IN	NP

Grammar

Vi	→	sleeps
Vt	→	saw
NN	→	man
NN	→	woman
NN	→	telescope
NN	→	dog
DT	→	the
IN	→	with
IN	→	in

Lexicon

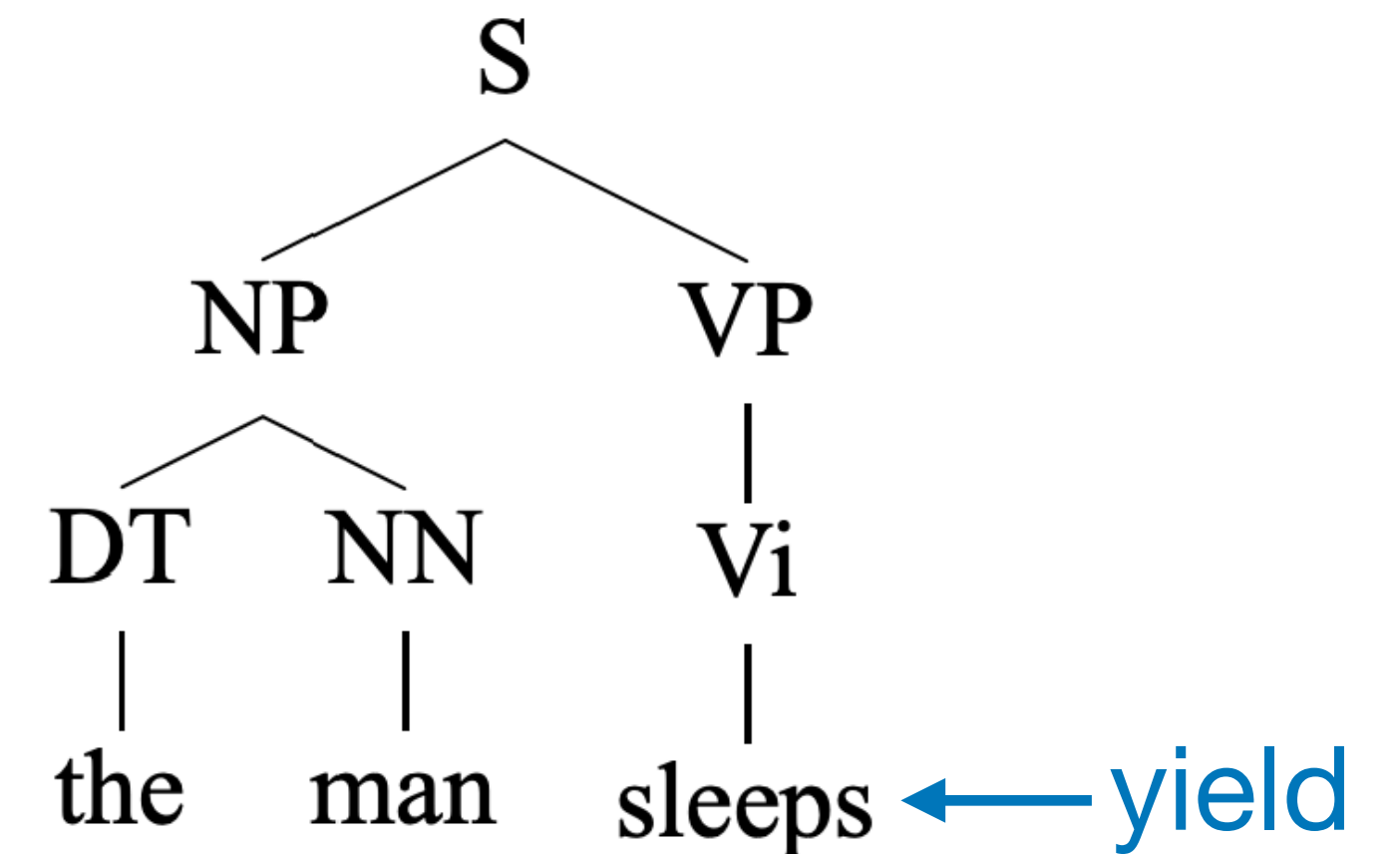
S:sentence, VP:verb phrase, NP: noun phrase, PP:prepositional phrase,
DT:determiner, Vi:intransitive verb, Vt:transitive verb, NN: noun, IN:preposition

Derivations

- Given a CFG G , a **derivation** is sequence of rule-expansions starting from the start symbol to a string consisting of terminal symbols
- It can be expressed as a sequence of strings $s_1, s_2, \dots, s_n$, where
 - $s_1 = S$ start symbol
 - $s_n \in \Sigma^*$ where Σ^* is all the possible strings made up of words from Σ
 - Each s_i for $i = 2, \dots, n$ is derived from s_{i-1} by picking some non-terminal X in s_{i-1} and replacing it by some β where $X \rightarrow \beta \in R$
- s_n : **yield** of the derivation

Left-most derivation:
pick left-most non-terminal

A derivation can be represented as a parse tree!

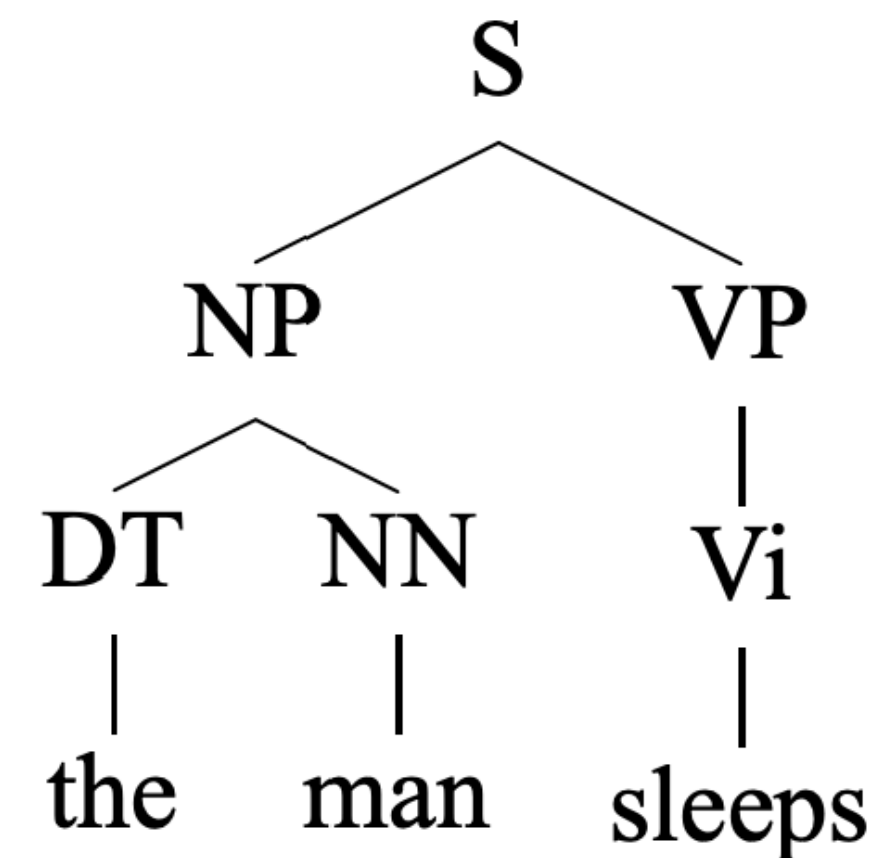


(Left-most) Derivation

- $s_1 = S$
- $s_2 = NP\ VP$
- $s_3 = DT\ NN\ VP$
- $s_4 = the\ NN\ VP$
- $s_5 = the\ man\ VP$
- $s_6 = the\ man\ Vi$
- $s_7 = the\ man\ sleeps$

$R =$

S	→	NP	VP
VP	→	Vi	
VP	→	Vt	NP
VP	→	VP	PP
NP	→	DT	NN
NP	→	NP	PP
PP	→	IN	NP



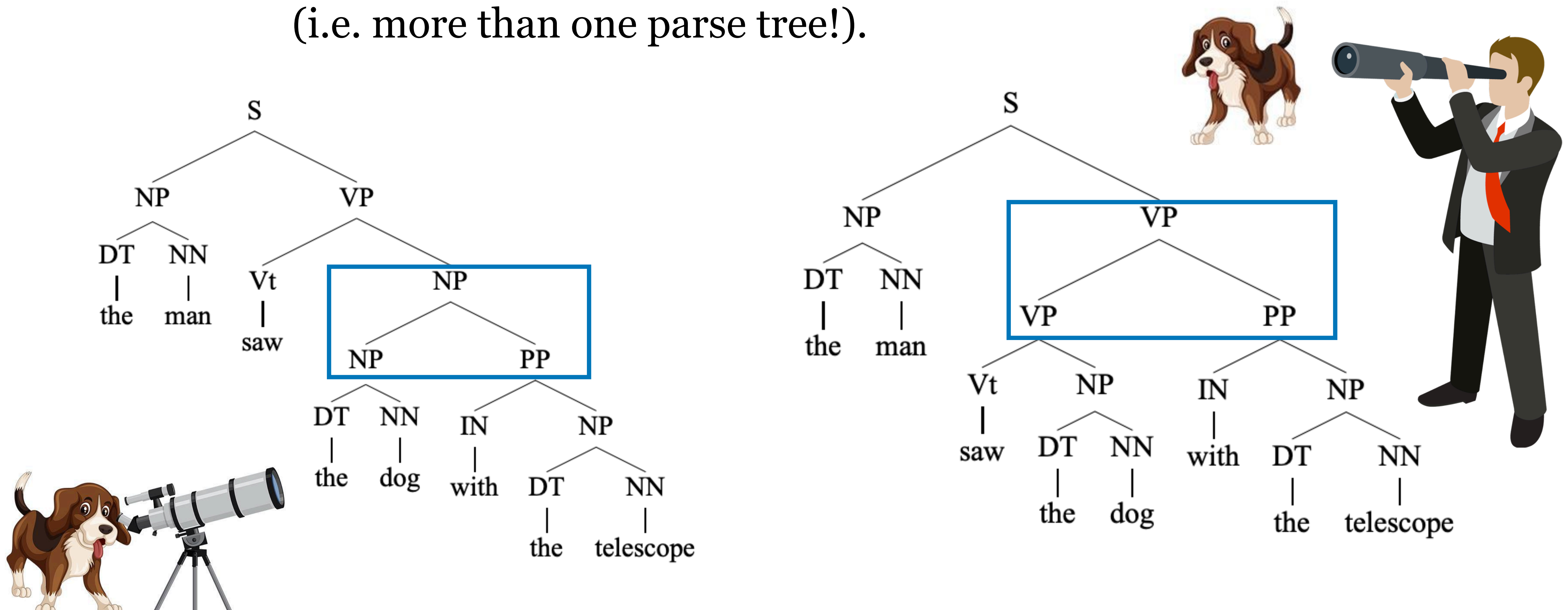
a parse tree

Vi	→	sleeps
Vt	→	saw
NN	→	man
NN	→	woman
NN	→	telescope
NN	→	dog
DT	→	the
IN	→	with
IN	→	in

- A string $s \in \Sigma^*$ is in the language defined by the CFG if there is at least one derivation whose yield is s
- The set of possible derivations may be finite or infinite

Ambiguity

- Some strings may have more than one derivations (i.e. more than one parse tree!).



“Classical” NLP Parsing

- In fact, sentences can have a **very large number of possible parses**

The board approved [its acquisition] [by Royal Trustco Ltd.] [of Toronto] [for \$27 a share] [at its monthly meeting].

- How many parses for sentence of length n ?

```
((natural language) (learning course))  
(((natural language) learning) course)  
((natural (language learning)) course)  
(natural (language (learning course)))  
(natural ((language learning) course))
```

$n : a^n$	number of parses
1	1
2	1
3	2
4	5
5	14
6	42
7	132
8	429
9	1430
10	4862
11	16796

“Classical” NLP Parsing

- In fact, sentences can have a **very large number of possible parses**

The board approved [its acquisition] [by Royal Trustco Ltd.] [of Toronto] [for \$27 a share] [at its monthly meeting].

- The number of (binary) parses happen to follow the Catalan numbers

$((ab)c)d$ $(a(bc))d$ $(ab)(cd)$ $a((bc)d)$ $a(b(cd))$

For a sentence of length n , can form constituents by placing parenthesis.

Number of parses = number of ways to parenthesize expression such that

- there are equal number of open/close parenthesis
- they are properly nested with open before close

Catalan number: $C_n = \frac{1}{n+1} \binom{2n}{n}$

unlabeled parses for a sentence of n words

$n : a^n$	number of parses
1	1
2	1
3	2
4	5
5	14
6	42
7	132
8	429
9	1430
10	4862
11	16796

See Church and Patil (CL Journal, 1982) or TAOCP VI pp 388-389 (Knuth, 1975)

“Classical” NLP Parsing

- In fact, sentences can have a **very large number of possible parses**
 - It is also **difficult to construct a grammar** with enough coverage
 - A less constrained grammar can parse more sentences but result in more parses for even simple sentences
 - There is no way to choose the right parse!
 - Need to be able to assign scores to parses
- ← Binary notion:
in or not in language

Statistical parsing

- Adding probabilities to the rules: probabilistic CFGs (PCFGs)
- Learning from data: treebanks

Treebanks: a collection of sentences paired with their parse trees

```
((S
  (NP-SBJ (DT That)
    (JJ cold) (, ,)
    (JJ empty) (NN sky) )
  (VP (VBD was)
    (ADJP-PRD (JJ full)
      (PP (IN of)
        (NP (NN fire)
          (CC and)
          (NN light) ))))
  (. .) ))
```

(a)

```
((S
  (NP-SBJ The/DT flight/NN )
  (VP should/MD
    (VP arrive/VB
      (PP-TMP at/IN
        (NP eleven/CD a.m/RB ))
      (NP-TMP tomorrow/NN )))))
```

(b)

Probabilistic context-free grammars (PCFGs)

- A CFG tells us whether a sentence is **in the language** it defines
- A PCFG gives us a mechanism for **assigning scores** (here, probabilities) to different parses for the same sentence.

Probabilistic context-free grammars (PCFGs)

S	→	NP	VP	1.0
VP	→	Vi		0.3
VP	→	Vt	NP	0.5
VP	→	VP	PP	0.2
NP	→	DT	NN	0.8
NP	→	NP	PP	0.2
PP	→	IN	NP	1.0

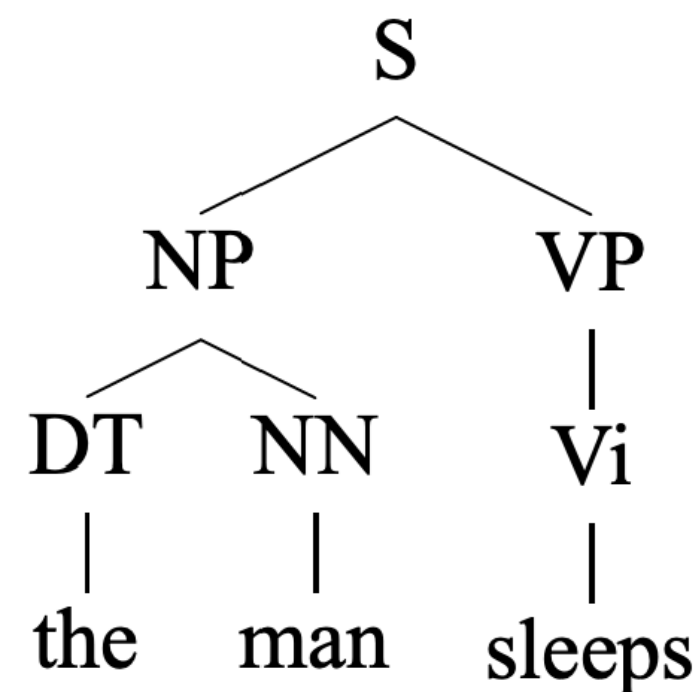
Vi	→	sleeps	1.0
Vt	→	saw	1.0
NN	→	man	0.1
NN	→	woman	0.1
NN	→	telescope	0.3
NN	→	dog	0.5
DT	→	the	1.0
IN	→	with	0.6
IN	→	in	0.4

- A probabilistic context-free grammar (PCFG) consists of:
 - A context-free grammar: $G = (N, \Sigma, R, S)$
 - For each rule $\alpha \rightarrow \beta \in R$, there is a parameter $q(\alpha \rightarrow \beta) \geq 0$.
For any $X \in N$,

$$\sum_{\alpha \rightarrow \beta: \alpha = X} q(\alpha \rightarrow \beta) = 1$$

Probabilistic context-free grammars (PCFGs)

For any derivation (parse tree) containing rules:
 $\alpha_1 \rightarrow \beta_1, \alpha_2 \rightarrow \beta_2, \dots, \alpha_l \rightarrow \beta_l$, the probability of the parse is: $\prod_{i=1}^l q(\alpha_i \rightarrow \beta_i)$



R				q
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VP	→	VP	PP	0.2
NP	→	DT	NN	0.8
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NN	→	dog	0.5
DT	→	the	1.0
IN	→	with	0.6
IN	→	in	0.4

$$\begin{aligned}
 P(t) &= q(S \rightarrow NP VP) \times q(NP \rightarrow DT NN) \times q(DT \rightarrow \text{the}) \\
 &\quad \times q(NN \rightarrow \text{man}) \times q(VP \rightarrow Vi) \times q(Vi \rightarrow \text{sleeps}) \\
 &= 1.0 \times 0.8 \times 1.0 \times 0.1 \times 0.3 \times 1.0 = 0.024
 \end{aligned}$$

Why do we want $\sum_{\alpha \rightarrow \beta: \alpha=X} q(\alpha \rightarrow \beta) = 1$?

Treebanks

English

- Standard setup (WSJ portion of Penn Treebank):
 - 40,000 sentences for training
 - 1,700 for development
 - 2,400 for testing
- Why building a treebank instead of a grammar?
 - Broad coverage
 - Frequencies and distributional information
 - A way to evaluate systems

Penn Treebank (1989-1996)

- Syntactic annotation of text for POS tagging, parses, predicate-arguments, and speech disfluencies
- WSJ articles from 3 years

Phrasal categories

Penn Treebank

ADJP	Adjective phrase
ADVP	Adverb phrase
NP	Noun phrase
PP	Prepositional phrase
S	Simple declarative clause
SBAR	Subordinate clause
SBARQ	Direct question introduced by <i>wh</i> -element
SINV	Declarative sentence with subject-aux inversion
SQ	Yes/no questions and subconstituent of SBARQ excluding <i>wh</i> -element
VP	Verb phrase
WHADVP	Wh-adverb phrase
WHNP	Wh-noun phrase
WHPP	Wh-prepositional phrase
X	Constituent of unknown or uncertain category
*	“Understood” subject of infinitive or imperative
0	Zero variant of <i>that</i> in subordinate clauses
T	Trace of wh-Constituent

Part-of-speech tagset

Penn Treebank

CC	Coordinating conj.	TO	infinitival <i>to</i>
CD	Cardinal number	UH	Interjection
DT	Determiner	VB	Verb, base form
EX	Existential there	VBD	Verb, past tense
FW	Foreign word	VBG	Verb, gerund/present pple
IN	Preposition	VCN	Verb, past participle
JJ	Adjective	VBP	Verb, non-3rd ps. sg. present
JJR	Adjective, comparative	VBZ	Verb, 3rd ps. sg. present
JJS	Adjective, superlative	WDT	Wh-determiner
LS	List item marker	WP	Wh-pronoun
MD	Modal	WP\$	Possessive <i>wh</i> -pronoun
NN	Noun, singular or mass	WRB	Wh-adverb
NNS	Noun, plural	#	Pound sign
NNP	Proper noun, singular	\$	Dollar sign
NNPS	Proper noun, plural	.	Sentence-final punctuation
PDT	Predeterminer	,	Comma
POS	Possessive ending	:	Colon, semi-colon
PRP	Personal pronoun	(	Left bracket character
PP\$	Possessive pronoun	)	Right bracket character
RB	Adverb	"	Straight double quote
RBR	Adverb, comparative	'	Left open single quote
RBS	Adverb, superlative	"	Left open double quote
RP	Particle	,	Right close single quote
SYM	Symbol	"	Right close double quote

Deriving a PCFG from a treebank

- Training data: a set of parse trees $t_1, t_2, \dots, t_m$
- A PCFG (N, Σ, S, R, q) :
 - N is the set of all **non-terminals** seen in the trees
 - Σ is the set of all **words** seen in the trees
 - S is taken to be the **start** symbol S .
 - R is taken to be the set of all **rules** $\alpha \rightarrow \beta$ seen in the trees
 - The maximum-likelihood parameter estimates are:

$$q_{ML}(\alpha \rightarrow \beta) = \frac{\text{Count}(\alpha \rightarrow \beta)}{\text{Count}(\alpha)} \quad \text{Can add smoothing}$$

If we have seen the rule $VP \rightarrow Vt NP$ 105 times, and the non-terminal VP 1000 times,
 $q(VP \rightarrow Vt NP) = 0.105$

What if there is no annotated parses?

- Use [Expectation Maximization](#).
- For learning parameters for PCFGs
 - E-Step: compute expectation over trees with fixed model weights (probabilities)
 - M-Step: determine model weights (probabilities) that maximize likelihood of expected parses
- Use the [inside-outside](#) algorithm (a dynamic programming algorithm) to compute these probabilities efficiently.

Parsing with PCFGs

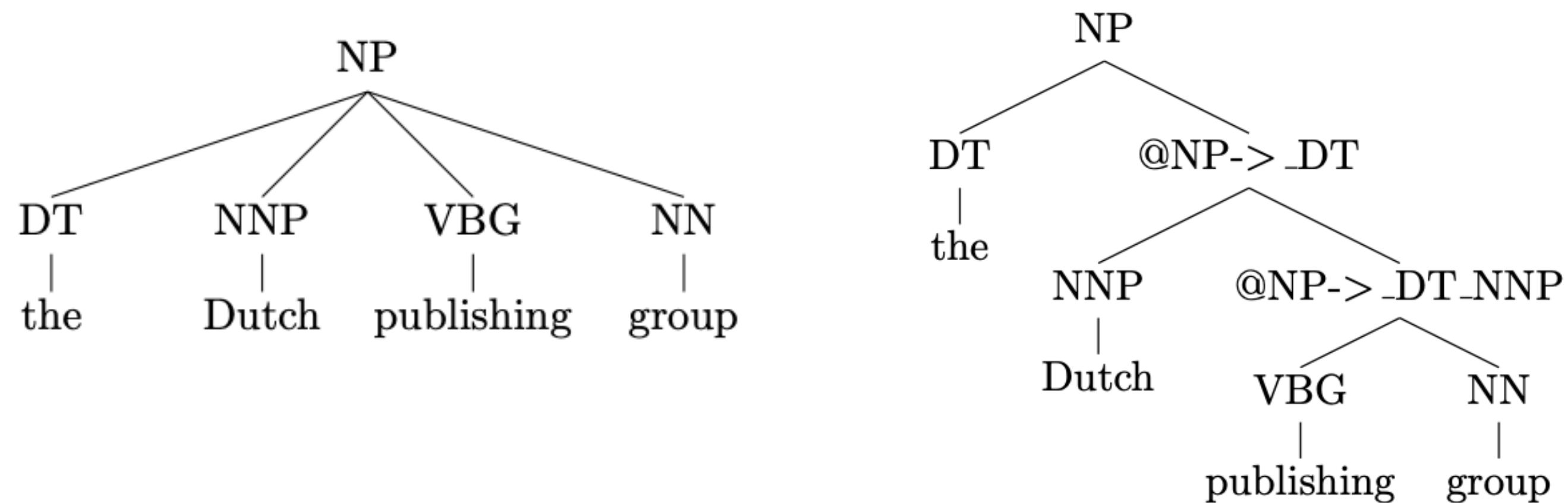
- Given a sentence s and a PCFG, how to find the **highest scoring parse tree** for s ?

$$\operatorname{argmax}_{t \in \mathcal{T}(s)} P(t)$$

- **The CKY algorithm:** applies to a PCFG in Chomsky normal form (CNF)
- **Chomsky Normal Form (CNF):** all the rules take one of the two following forms:
 - $X \rightarrow Y_1 Y_2$ where $X \in N, Y_1 \in N, Y_2 \in N$ **Binary**
 - $X \rightarrow Y$ where $X \in N, Y \in \Sigma$ **Unary**
- Can convert any PCFG into an equivalent grammar in CNF!
 - However, the trees will look differently
 - Possible to do “reverse transformation”

Converting PCFGs into a CNF grammar

- n -ary rules ($n > 2$): $\text{NP} \rightarrow \text{DT NNP VBG NN}$



- Unary rules: $\text{VP} \rightarrow \text{Vi}$, $\text{Vi} \rightarrow \text{sleeps}$
 - Eliminate all the unary rules recursively by adding $\text{VP} \rightarrow \text{sleeps}$
 - We will come back to this later!

The CKY algorithm

Cocke–Kasami–Younger

- Dynamic programming
- Given a sentence $x_1, x_2, \dots, x_n$, denote $\pi(i, j, X)$ as the highest score for any parse tree that dominates words $x_i, \dots, x_j$ and has non-terminal $X \in N$ as its root.
- Output: $\pi(1, n, S)$
- Initially, for $i = 1, 2, \dots, n$,

$$\pi(i, i, X) = \begin{cases} q(X \rightarrow x_i) & \text{if } X \rightarrow x_i \in R \\ 0 & \text{otherwise} \end{cases}$$

Book the flight through Houston

1 2 3 4 5

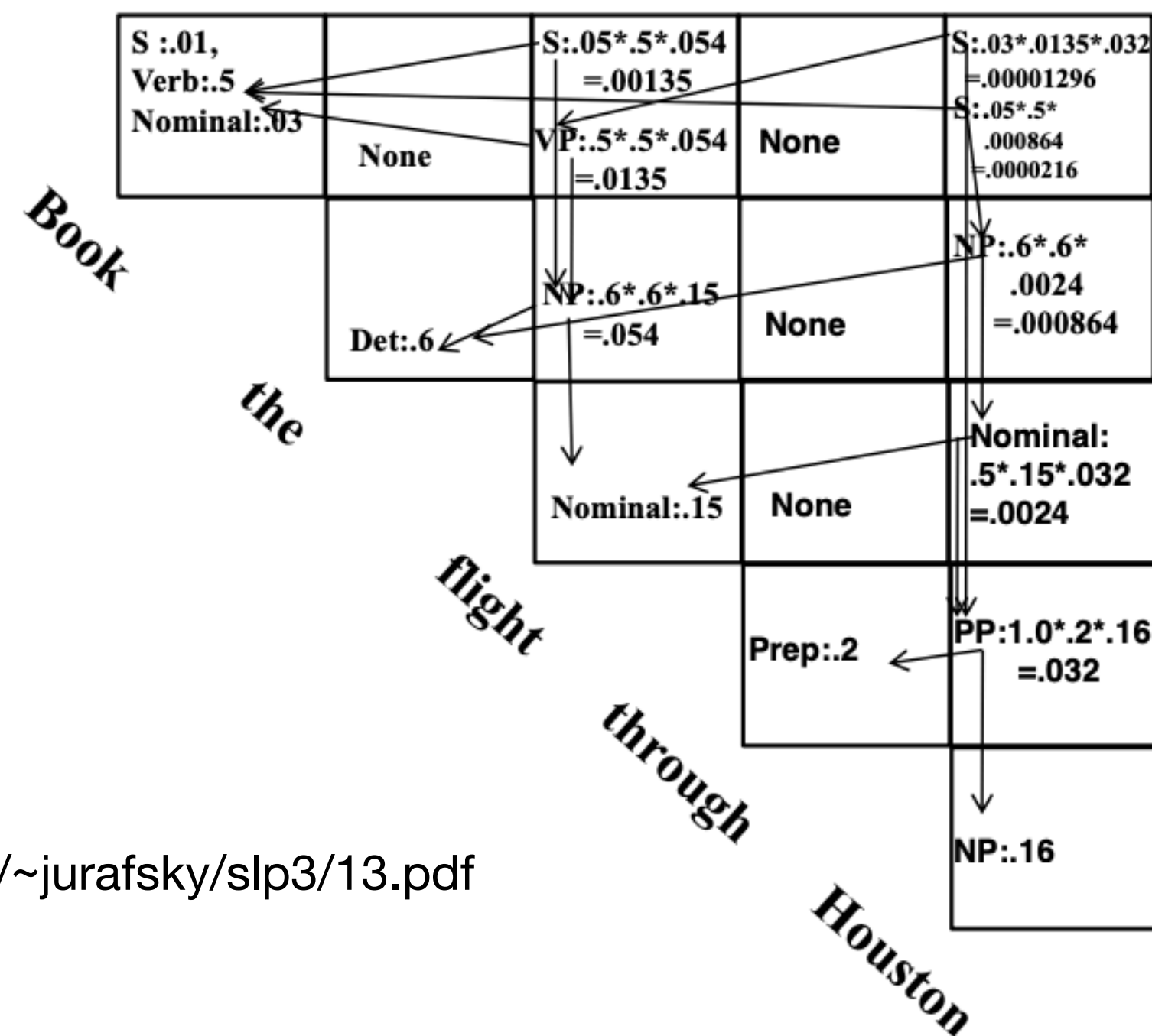
The CKY algorithm

- For all (i, j) such that $1 \leq i < j \leq n$ for all $X \in N$,

$$\pi(i, j, X) = \max_{X \rightarrow YZ \in R, i \leq k < j} q(X \rightarrow YZ) \times \pi(i, k, Y) \times \pi(k + 1, j, Z)$$

Consider all ways span (i,j) can be split into 2 (k is the split point)

Also stores backpointers which allow us to recover the parse tree



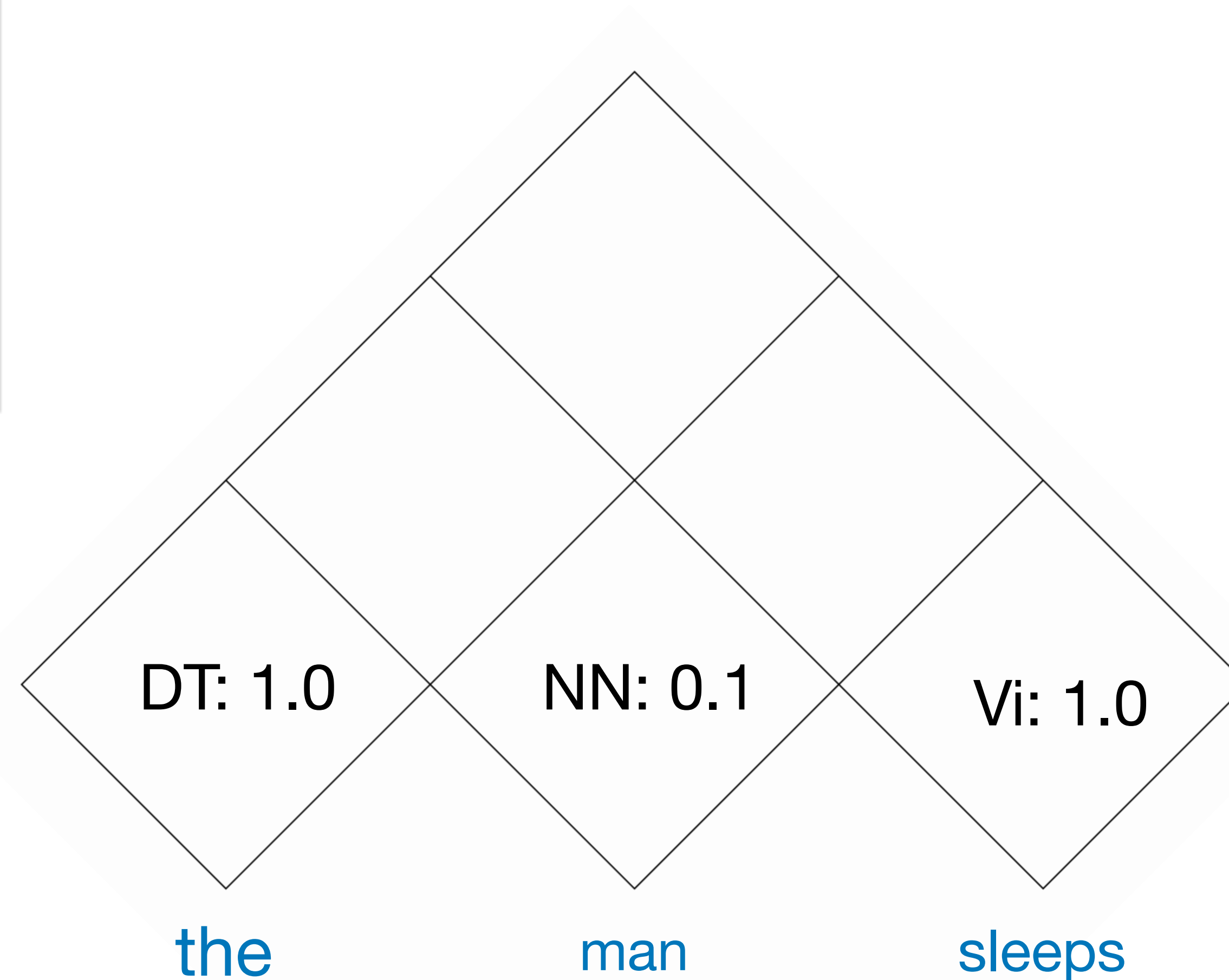
Cells contain:

- Best score for parse of span (i,j) for each non-terminal X
- Backpointers

Example of CKY parsing

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VP	→	VP	PP	0.2
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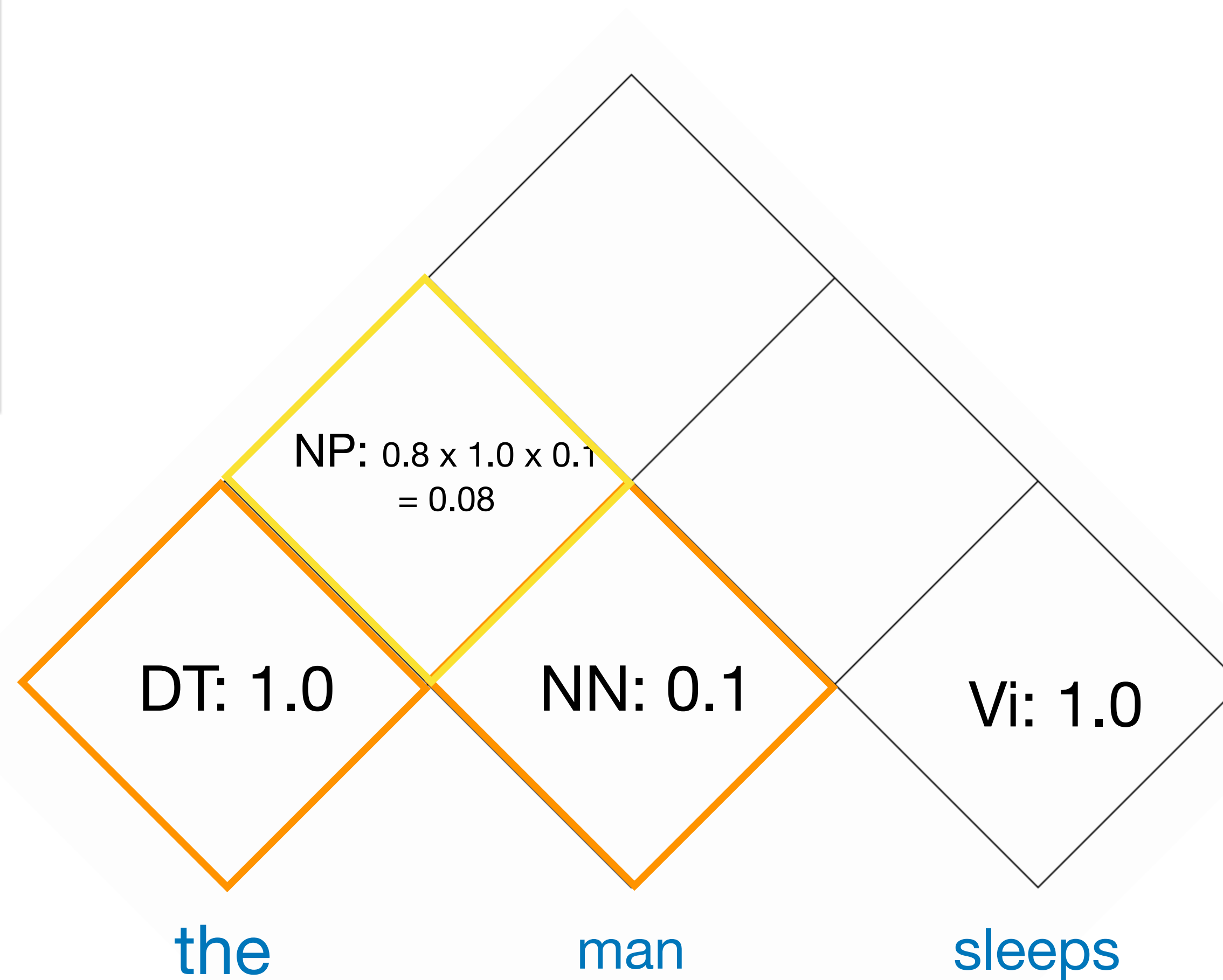
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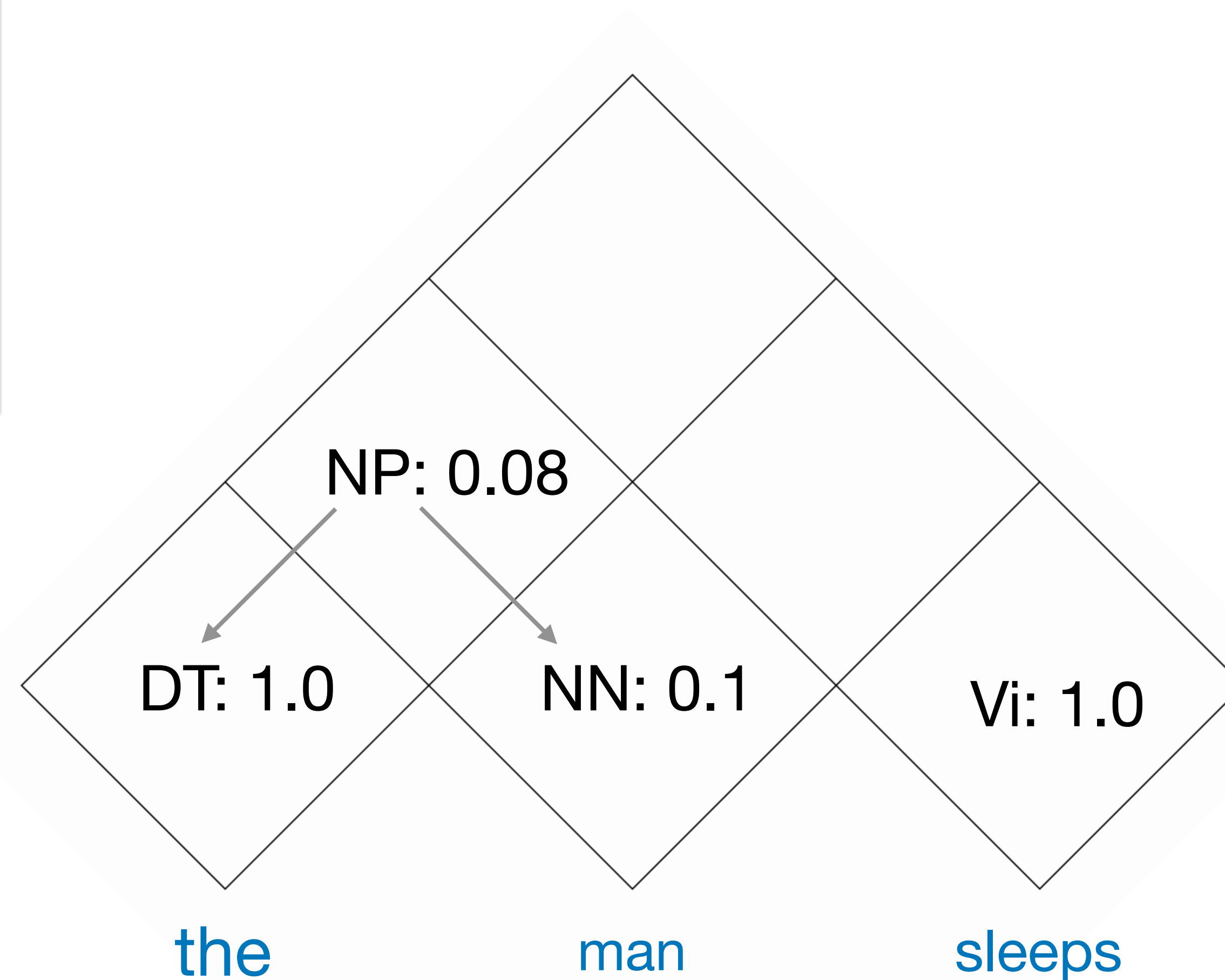
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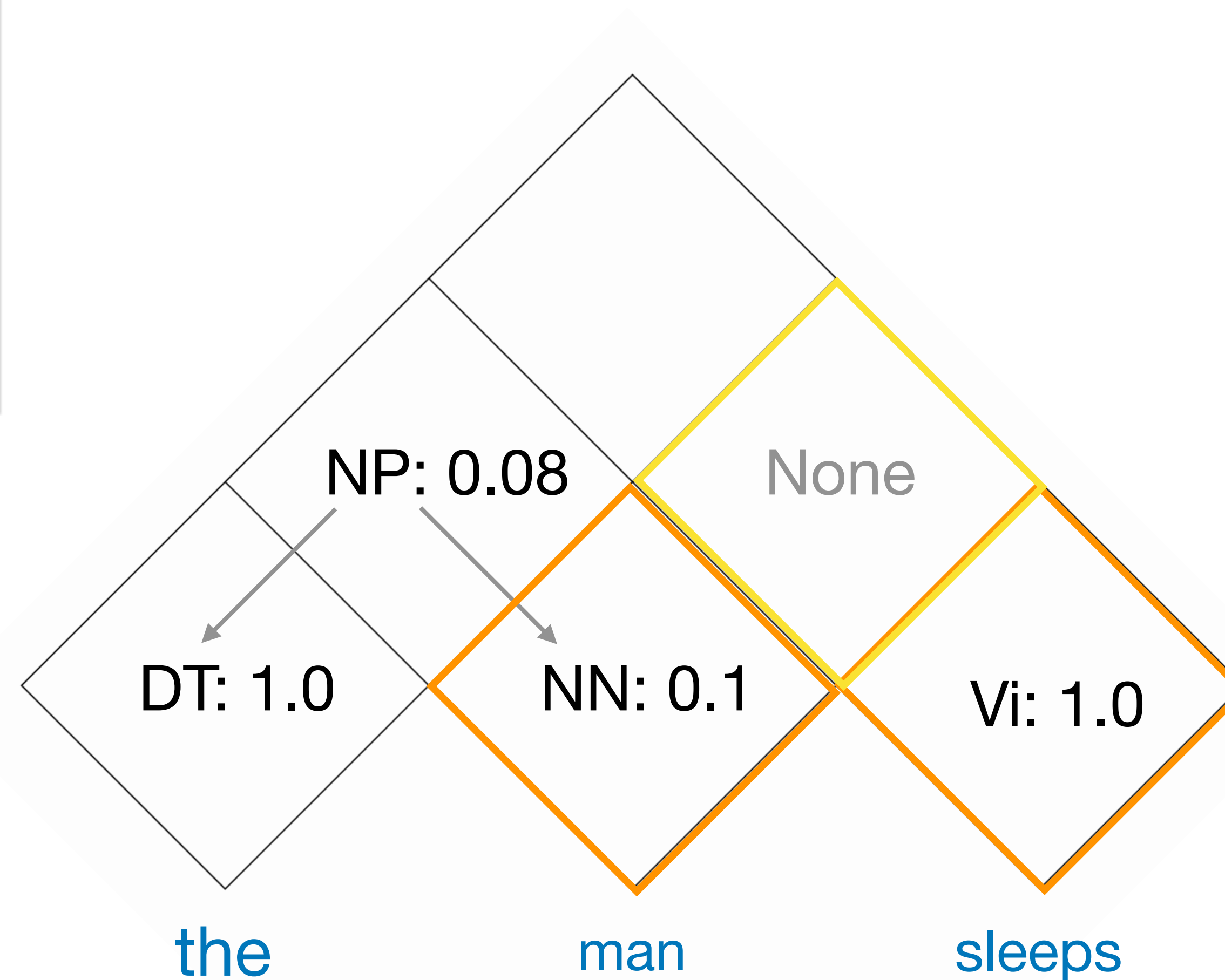
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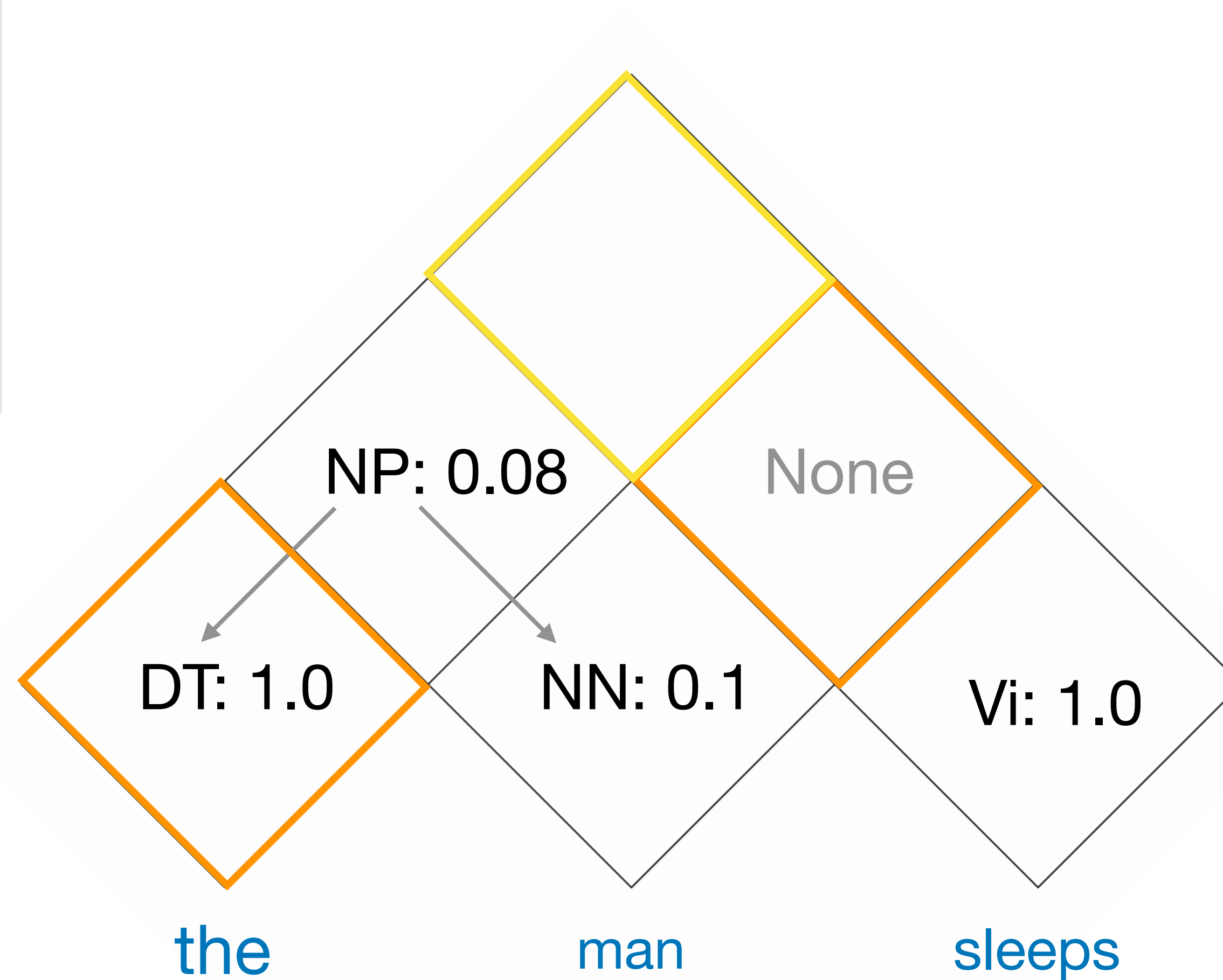
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NP	→	NP	PP	0.2
PP	→	IN	NP	1.0

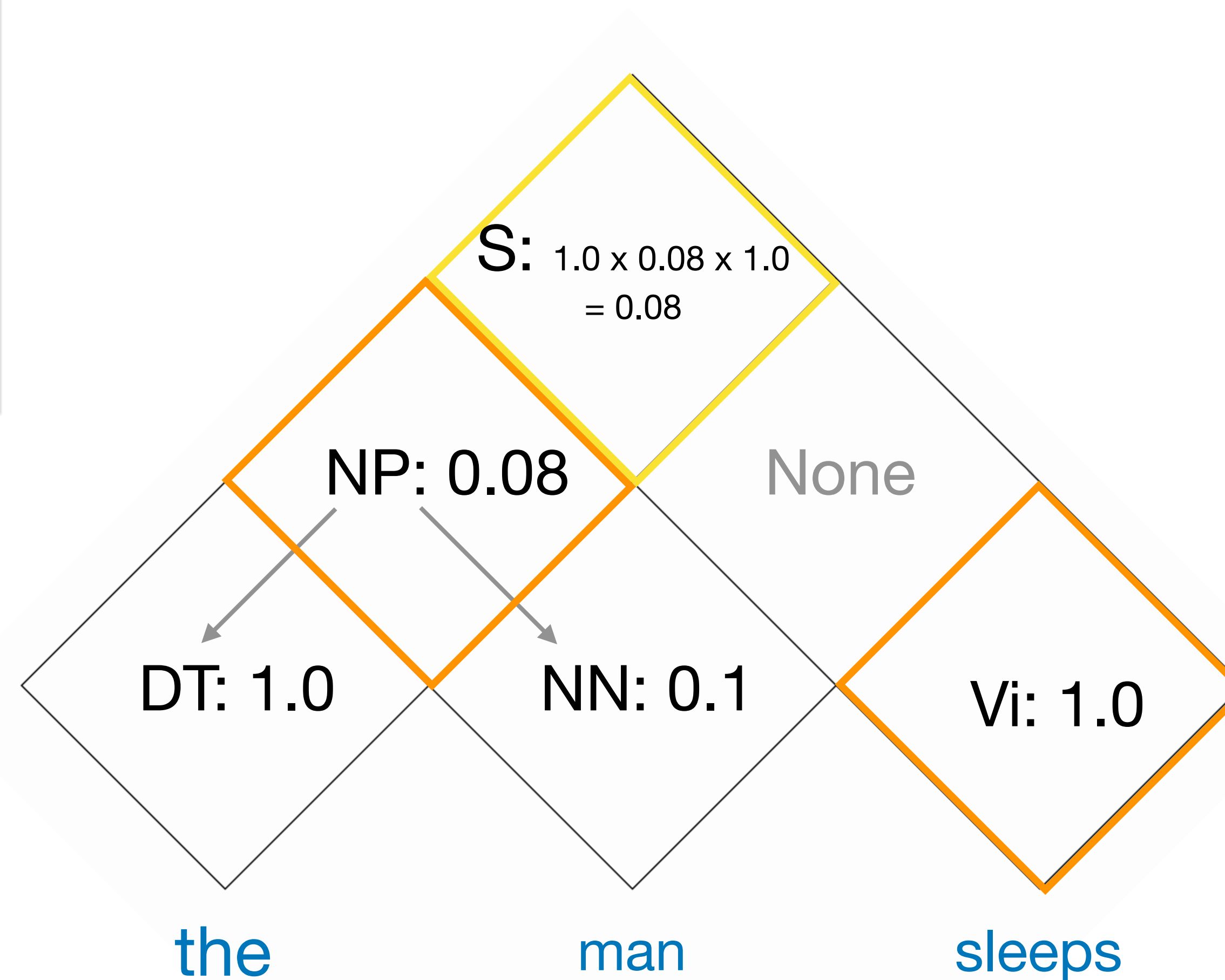
Vi	→	sleeps	1.0
Vt	→	saw	1.0
NN	→	man	0.1
NN	→	woman	0.1
NN	→	telescope	0.3
NN	→	dog	0.5
DT	→	the	1.0
IN	→	with	0.6
IN	→	in	0.4



Example of CKY parsing

S	→	NP	VP	1.0
VP	→	Vi		0.3
VP	→	Vt	NP	0.5
VP	→	VP	PP	0.2
NP	→	DT	NN	0.8
NP	→	NP	PP	0.2
PP	→	IN	NP	1.0

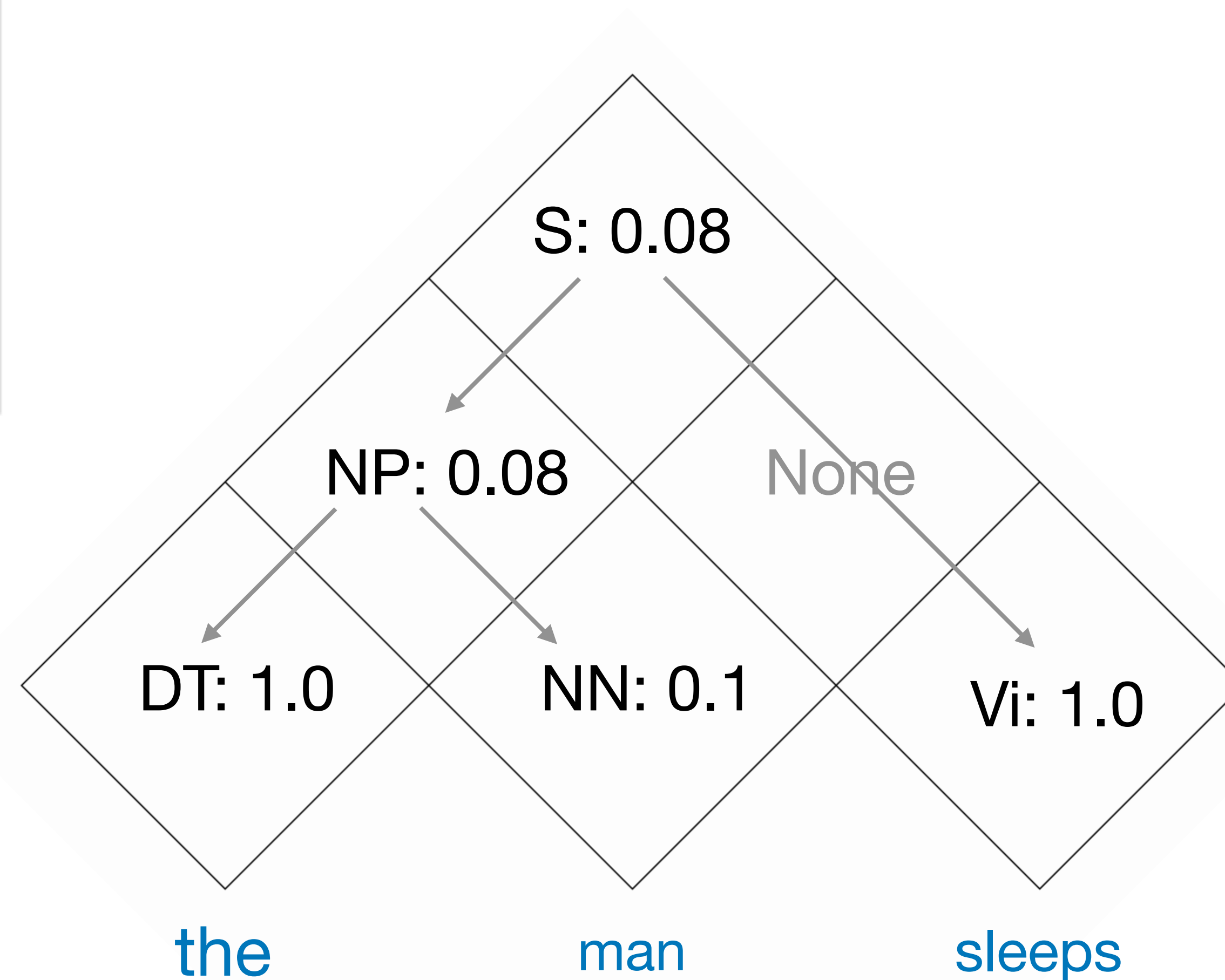
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Example of CKY parsing

S	→	NP	VP	1.0
VP	→	Vi		0.3
VP	→	Vt	NP	0.5
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NP	→	DT	NN	0.8
NP	→	NP	PP	0.2
PP	→	IN	NP	1.0

Vi	→	sleeps	1.0
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IN	→	in	0.4



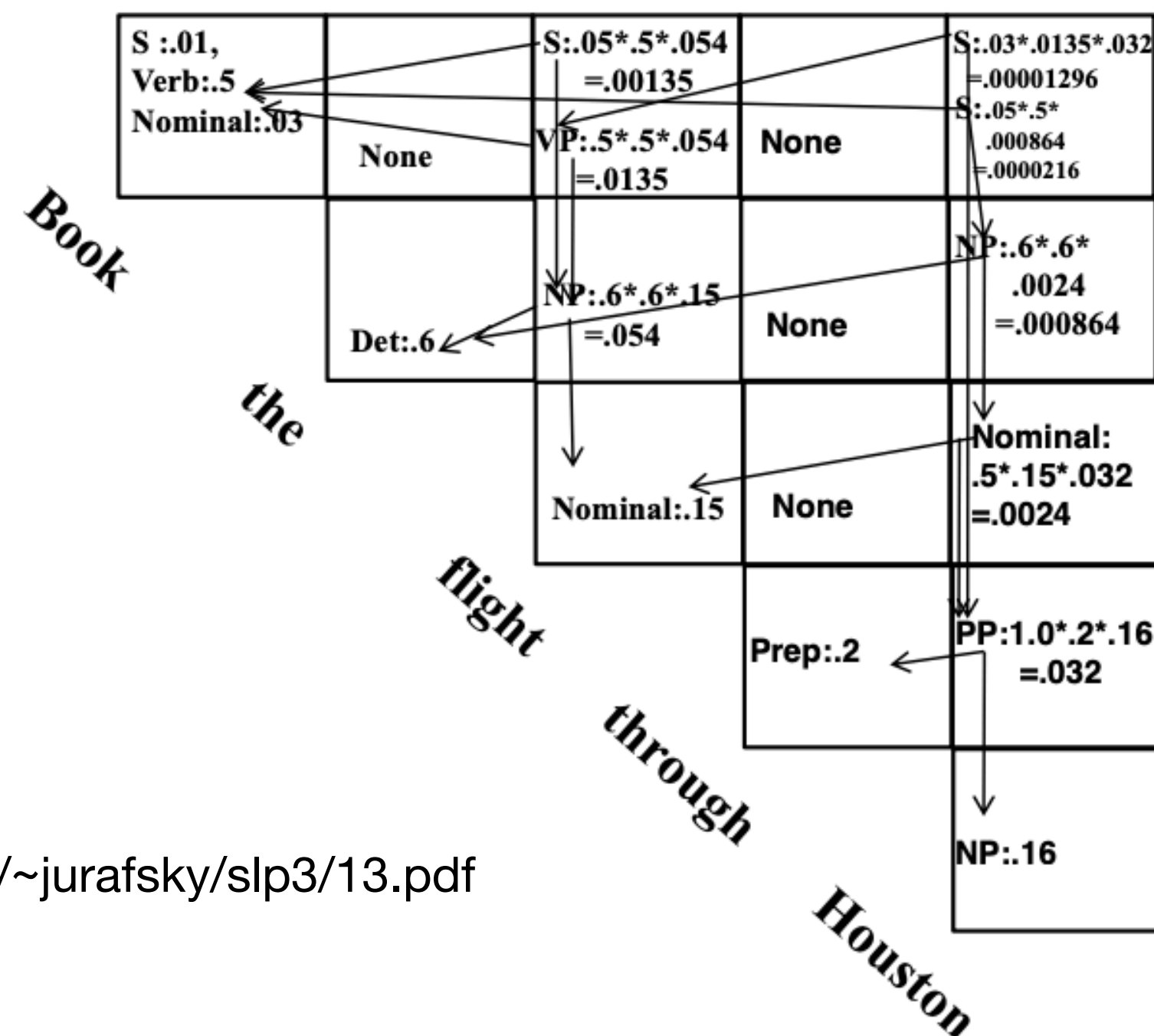
The CKY algorithm

- For all (i, j) such that $1 \leq i < j \leq n$ for all $X \in N$,

$$\pi(i, j, X) = \max_{X \rightarrow YZ \in R, i \leq k < j} q(X \rightarrow YZ) \times \pi(i, k, Y) \times \pi(k + 1, j, Z)$$

Consider all ways span (i,j) can be split
into 2 (k is the split point)

Also stores backpointers which allow us to recover the parse tree



Cells contain:

- Best score for parse of span (i,j) for each non-terminal X
- Backpointers

The CKY algorithm

Input: a sentence $s = x_1 \dots x_n$, a PCFG $G = (N, \Sigma, S, R, q)$.

Initialization:

For all $i \in \{1 \dots n\}$, for all $X \in N$,

$$\pi(i, i, X) = \begin{cases} q(X \rightarrow x_i) & \text{if } X \rightarrow x_i \in R \\ 0 & \text{otherwise} \end{cases}$$

Algorithm:

- For $l = 1 \dots (n - 1)$
 - For $i = 1 \dots (n - l)$
 - * Set $j = i + l$
 - * For all $X \in N$, calculate

$$\pi(i, j, X) = \max_{\substack{X \rightarrow YZ \in R, \\ s \in \{i \dots (j-1)\}}} (q(X \rightarrow YZ) \times \pi(i, s, Y) \times \pi(s + 1, j, Z))$$

and

$$bp(i, j, X) = \arg \max_{\substack{X \rightarrow YZ \in R, \\ s \in \{i \dots (j-1)\}}} (q(X \rightarrow YZ) \times \pi(i, s, Y) \times \pi(s + 1, j, Z))$$

Output: Return $\pi(1, n, S) = \max_{t \in \mathcal{T}(s)} p(t)$, and backpointers bp which allow recovery of $\arg \max_{t \in \mathcal{T}(s)} p(t)$.

Running time?

$$O(n^3 |R|)$$

CKY with unary rules

- In practice, we also allow unary rules:

$$X \rightarrow Y \text{ where } X, Y \in N$$

conversion to/from the normal form is easier

$$\pi(i, j, X) = \max_{X \rightarrow Y \in R} q(X \rightarrow Y) \times \pi(i, j, Y)$$

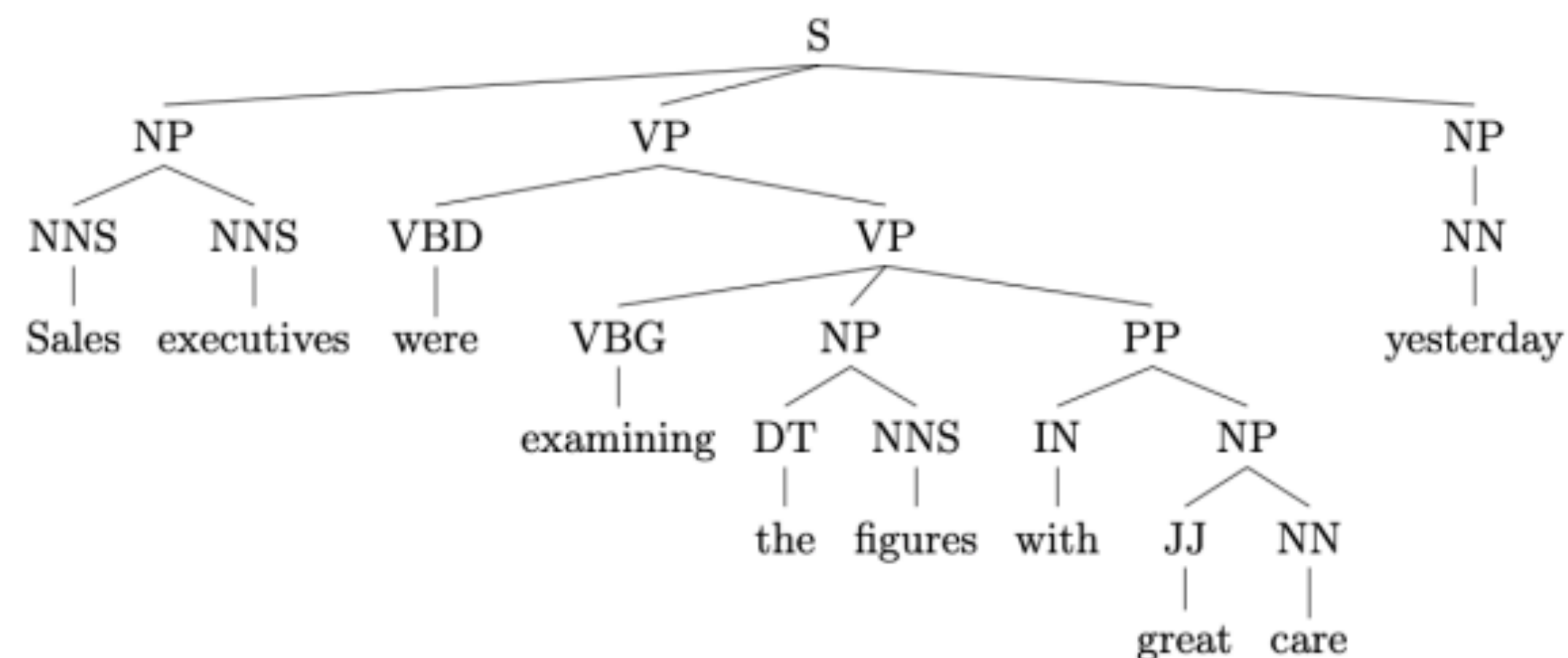
- Compute unary closure: if there is a rule chain $X \rightarrow Y_1, Y_1 \rightarrow Y_2, \dots, Y_k \rightarrow Y$, add $q(X \rightarrow Y) = q(X \rightarrow Y_1) \times \dots \times q(Y_k \rightarrow Y)$
- Update unary rule once after the binary rules

Constituency Parsing

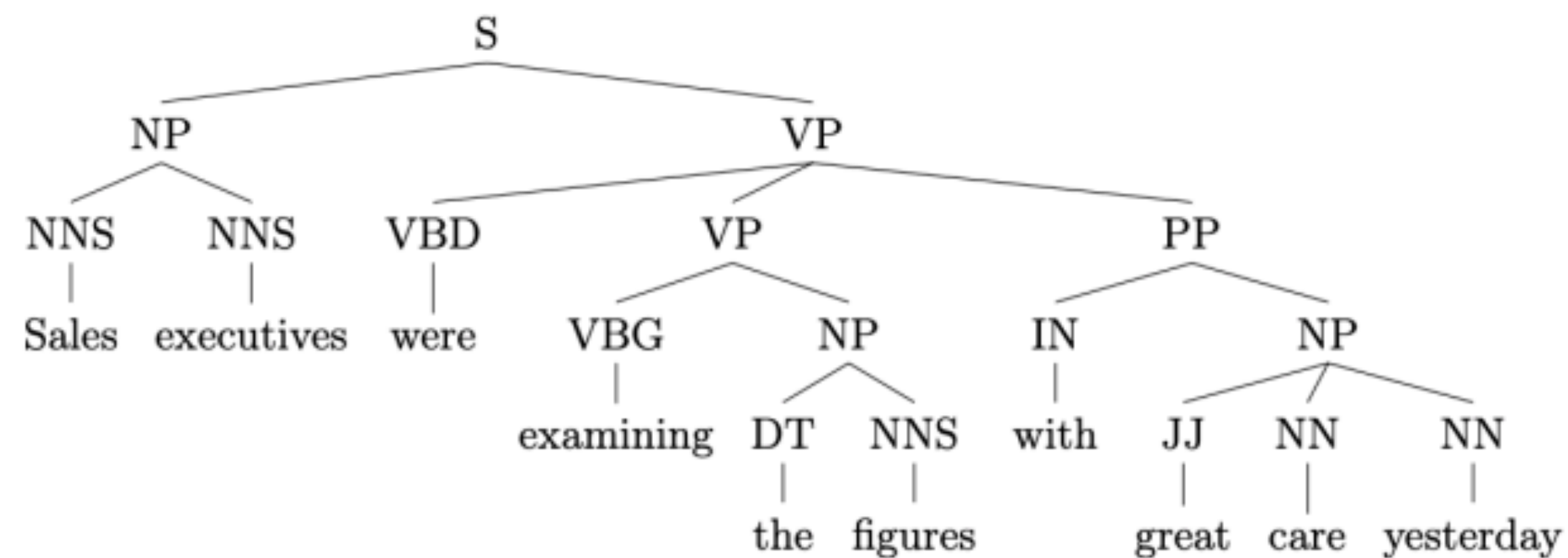
- Borealis AI Tutorials
 - Parsing I (<https://www.borealisai.com/en/blog/tutorial-15-parsing-i-context-free-grammars-and-cyk-algorithm/>)
 - CFGs and the CKY algorithm
 - CNF and number of parses
 - Parsing II (<https://www.borealisai.com/en/blog/tutorial-18-parsing-ii-wcfgs-inside-algorithm-and-weighted-parsing/>)
 - Weighted CFGs and CKY algorithm for parsing Weighted CFGs
 - Parsing III (<https://www.borealisai.com/en/blog/tutorial-19-parsing-iii-pcfgs-and-inside-outside-algorithm/>)
 - PCFGs
 - Parameter estimation for both supervised and unsupervised cases
 - Inside-Outside algorithm for unsupervised learning of parameters

Evaluating constituency parsing

Gold: (1, 10, S), (1, 2, NP), (3, 9, VP), (4, 9, VP), (5, 6, NP), (7, 9, PP), (8, 9, NP), (10, 10, NP)



Predicted: (1, 10, S), (1, 2, NP), (3, 10, VP), (4, 6, VP), (5, 6, NP), (7, 10, PP), (8, 10, NP)

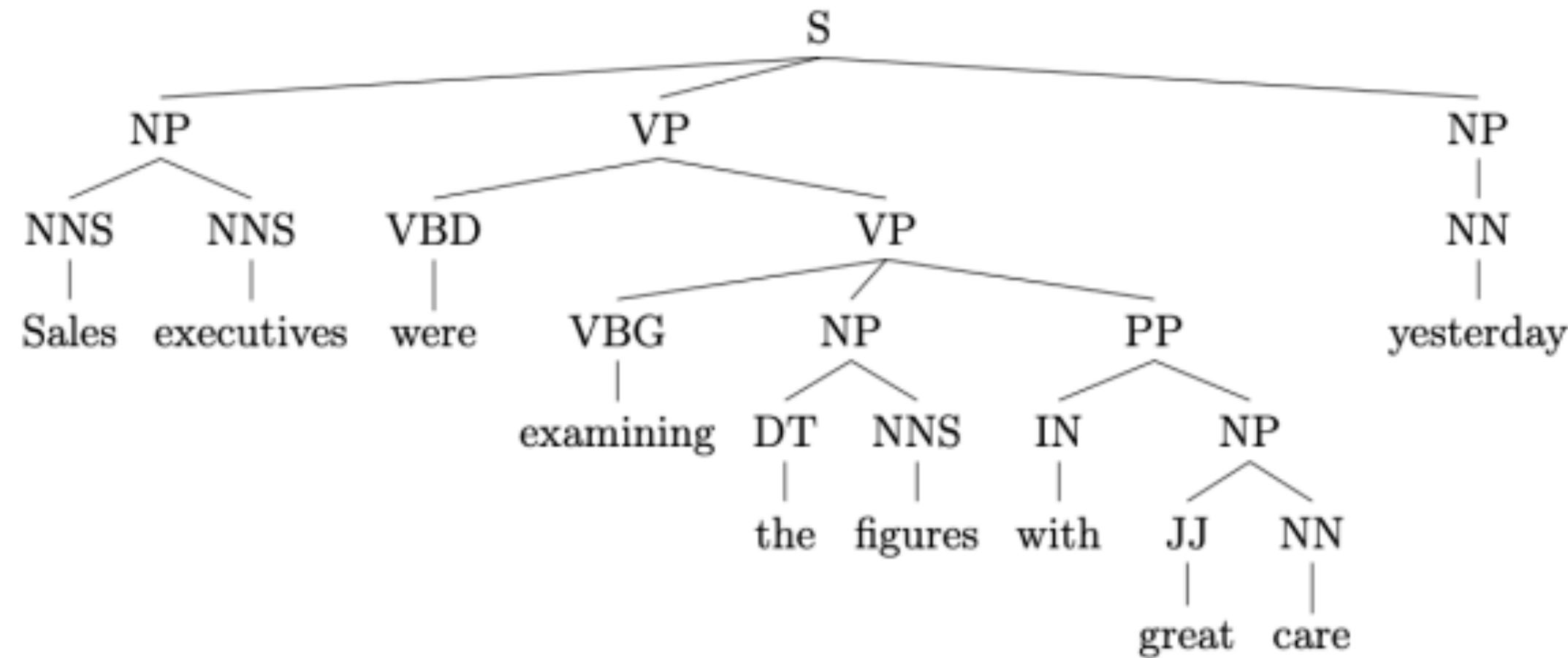


Evaluating constituency parsing

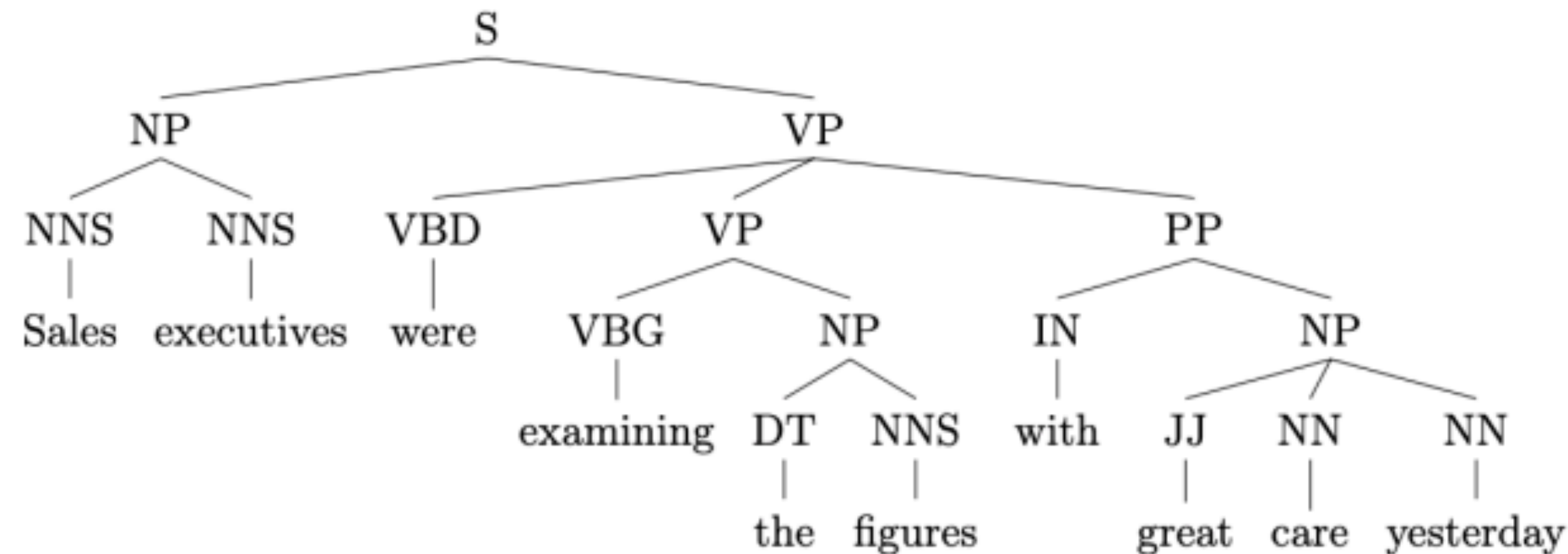
- Recall: $(\# \text{ correct constituents in candidate}) / (\# \text{ constituents in gold tree})$
- Precision: $(\# \text{ correct constituents in candidate}) / (\# \text{ constituents in candidate})$
- Labeled precision/recall require getting the non-terminal label correct
- $F1 = (2 * \text{precision} * \text{recall}) / (\text{precision} + \text{recall})$
- Part-of-speech tagging accuracy is evaluated separately

Evaluating constituency parsing

Gold: (1, 10, S), (1, 2, NP), (3, 9, VP), (4, 9, VP), (5, 6, NP), (7, 9, PP), (8, 9, NP), (10, 10, NP)



Predicted: (1, 10, S), (1, 2, NP), (3, 10, VP), (4, 6, VP), (5, 6, NP), (7, 10, PP), (8, 10, NP)



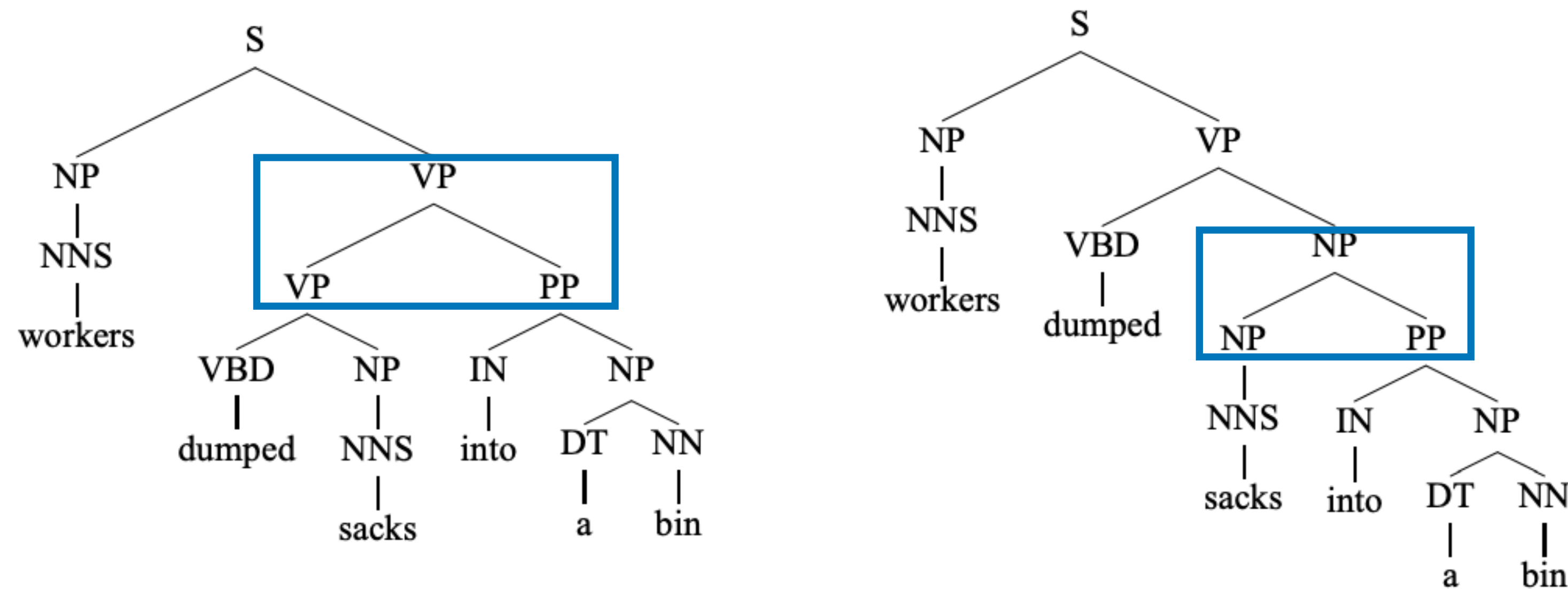
- Precision: $3/7 = 42.9\%$
- Recall: $3/8 = 37.5\%$
- F1 = 40.0%
- Tagging accuracy: 100%

Weaknesses of PCFGs

- Strong independence assumption
 - Each production (e.g., NP $\rightarrow$ DT NN) is **independent** of the rest of the tree
- Lack of sensitivity to context (where is the non-terminal in the tree, is it a subject or object)
- Lack of sensitivity to lexical information (words)

Weaknesses of PCFGs

- Lack of sensitivity to lexical information (words)



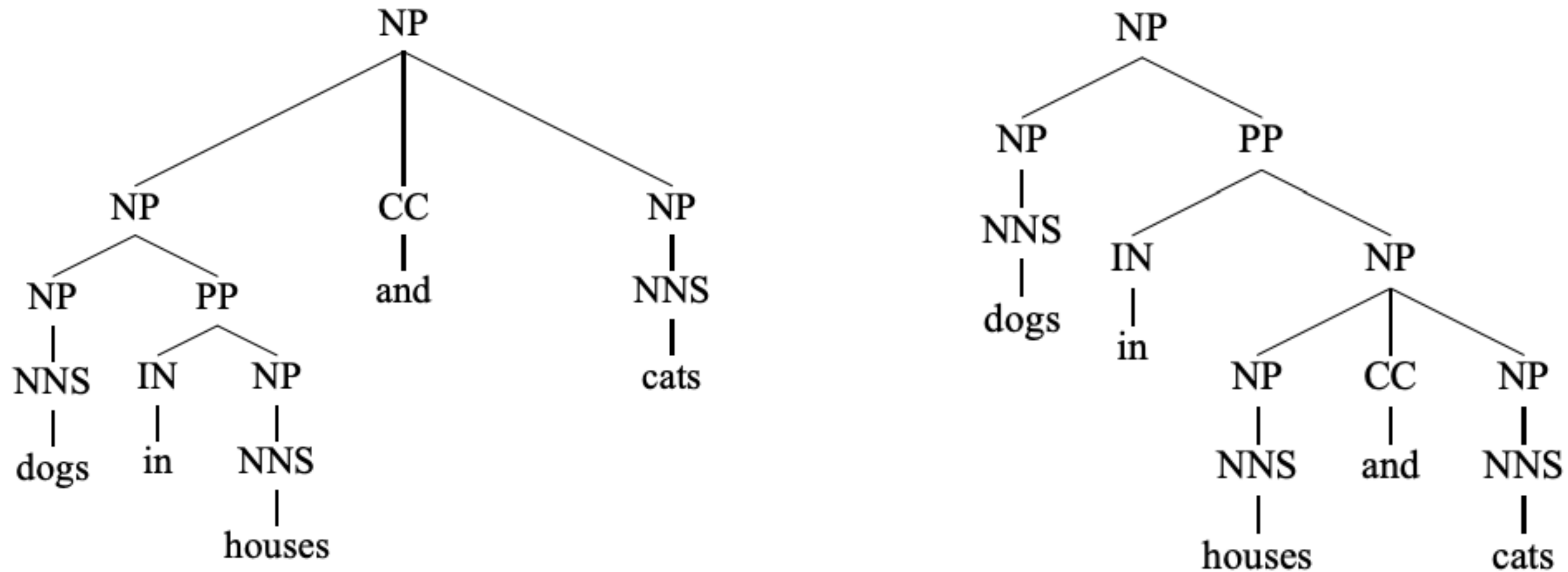
The only difference between these two parses:

$$q(\text{VP} \rightarrow \text{VP PP}) \text{ vs } q(\text{NP} \rightarrow \text{NP PP})$$

Difficult to determine the correct parse without looking at the words!

Weaknesses of PCFGs

- Lack of sensitivity to lexical information (words)

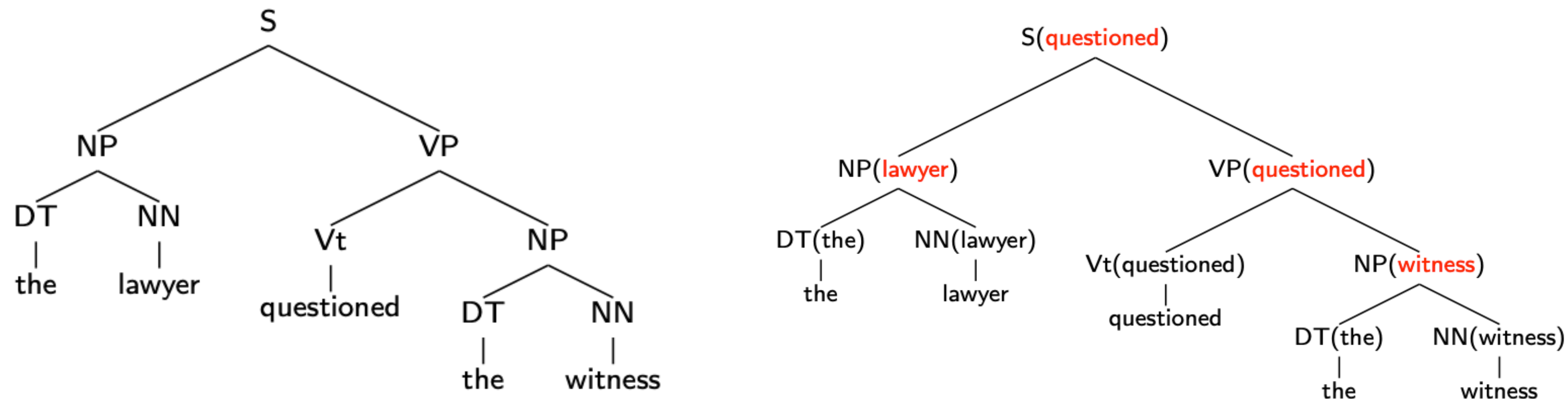


Exactly the same set of context-free rules!

Lexicalized PCFGs

- Key idea: add **headwords** to trees

Annotate parent with more information



- Each context-free rule has one special child that is the head of the rule (a core idea in syntax)

S	⇒	NP	VP	(VP is the head)
VP	⇒	Vt	NP	(Vt is the head)
NP	⇒	DT	NN	NN (NN is the head)

Head finding rules

If the rule contains NN, NNS, or NNP:

Choose the rightmost NN, NNS, or NNP

Else If the rule contains an NP: Choose the leftmost NP

Else If the rule contains a JJ: Choose the rightmost JJ

Else If the rule contains a CD: Choose the rightmost CD

Else Choose the rightmost child

If the rule contains Vi or Vt: Choose the leftmost Vi or Vt

Else If the rule contains a VP: Choose the leftmost VP

Else Choose the leftmost child

Lexicalized PCFGs

S(saw)	$\rightarrow_2$	NP(man)	VP(saw)
VP(saw)	$\rightarrow_1$	Vt(saw)	NP(dog)
NP(man)	$\rightarrow_2$	DT(the)	NN(man)
NP(dog)	$\rightarrow_2$	DT(the)	NN(dog)
Vt(saw)	$\rightarrow$	saw	
DT(the)	$\rightarrow$	the	
NN(man)	$\rightarrow$	man	
NN(dog)	$\rightarrow$	dog	

Drawbacks:

- Dramatically increases the size of the grammar -> less training data for each production
- Increase the complexity of the model (running time and memory)

- Further reading: *Michael Collins. 2003. Head-Driven Statistical Models for Natural Language Parsing.*
- Results for a PCFG: 70.6% recall, 74.8% precision
- Results for a lexicalized PCFG: 88.1% recall, 88.3% precision

Further improvements to parsing

- Discriminative **reranking**
 - PCFG is a generative model
 - Use discriminative models with more global features to score parses and rerank candidate parses from the PCFG
- **Self-training** (incorporate unlabeled data)
 - Train on some data to get initial good model
 - Then run model on unlabeled data and combine newly labeled data with gold labeled data and retrain
- **Ensemble**
 - Combine multiple models

Charniak parser w/
self-train+rerank:
(McClosky et al 2006)
92.1 F1

Beyond supervised learning:

Grammar Induction = learn grammar from unlabeled data

Using Neural Networks for Constituency Parsing

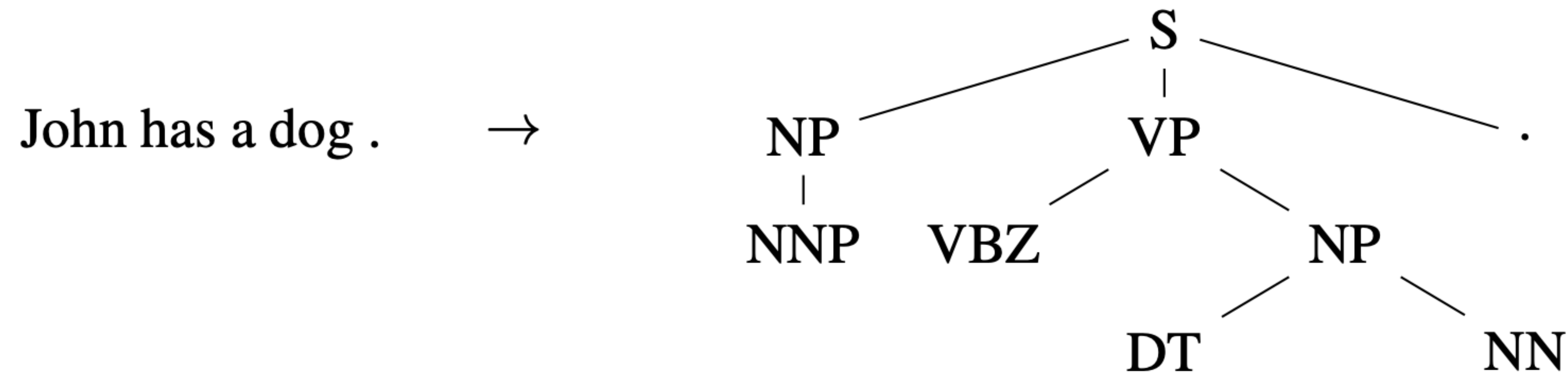
Parsing with Neural Networks

What can neural networks bring?

- Better phrase representations
 - Embeddings for words, tags, and nodes
 - Leverage pretrained embeddings
- Learned scoring functions
- Less independence assumptions

Parsing as Seq2Seq

(Vinyals et al, 2015; Vaswani et al, 2017)



John has a dog . → (S (NP NNP)_{NP} (VP VBZ (NP DT NN)_{NP})_{VP} .)_S

May not be structural correct
(i.e. unbalanced parenthesis)

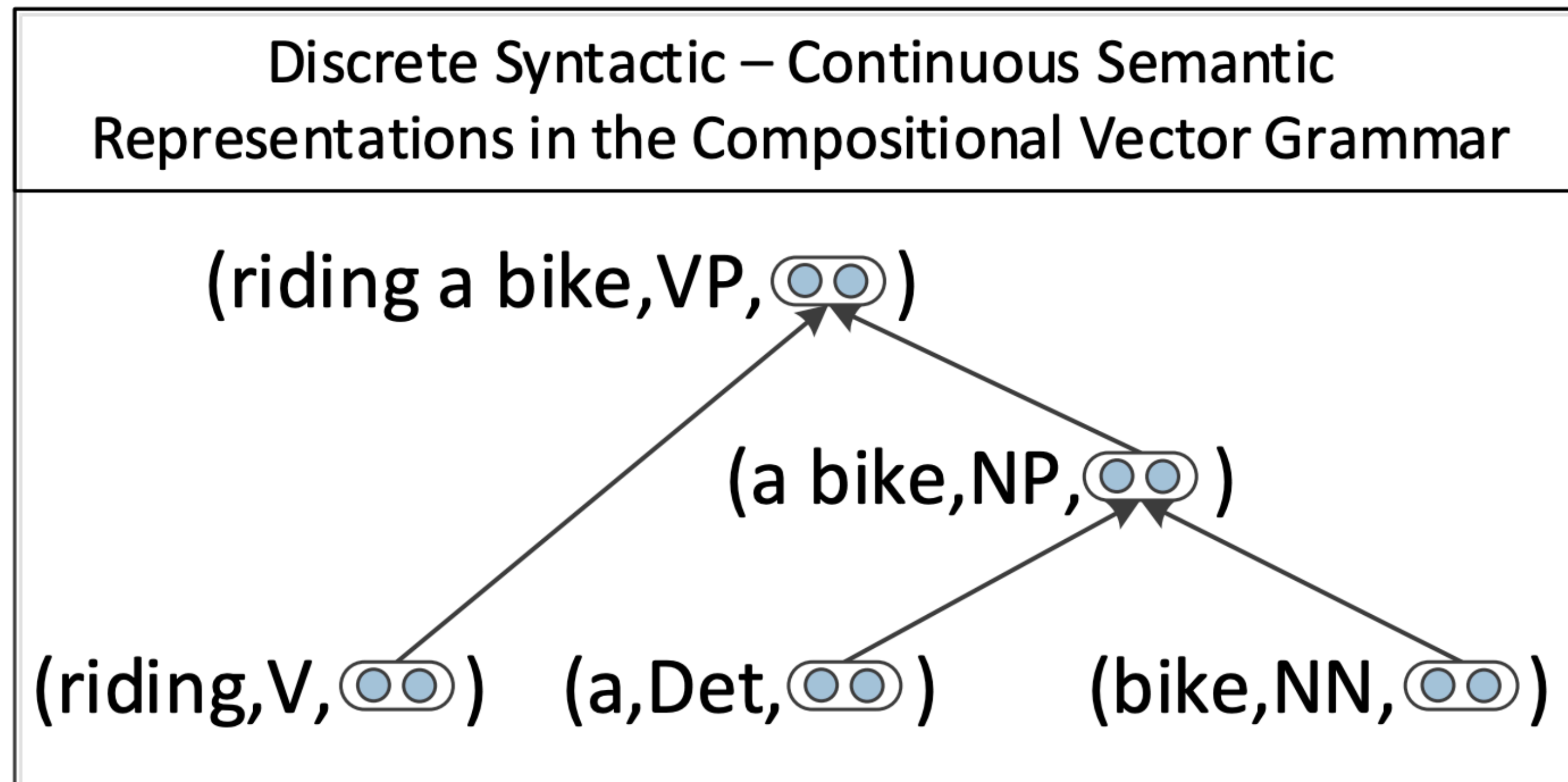
88.3 F1

91.3 F1

- Linearize parse tree and train LSTM seq2seq model with attention
- With transformers

Recursive Neural Networks (Socher et al, 2013)

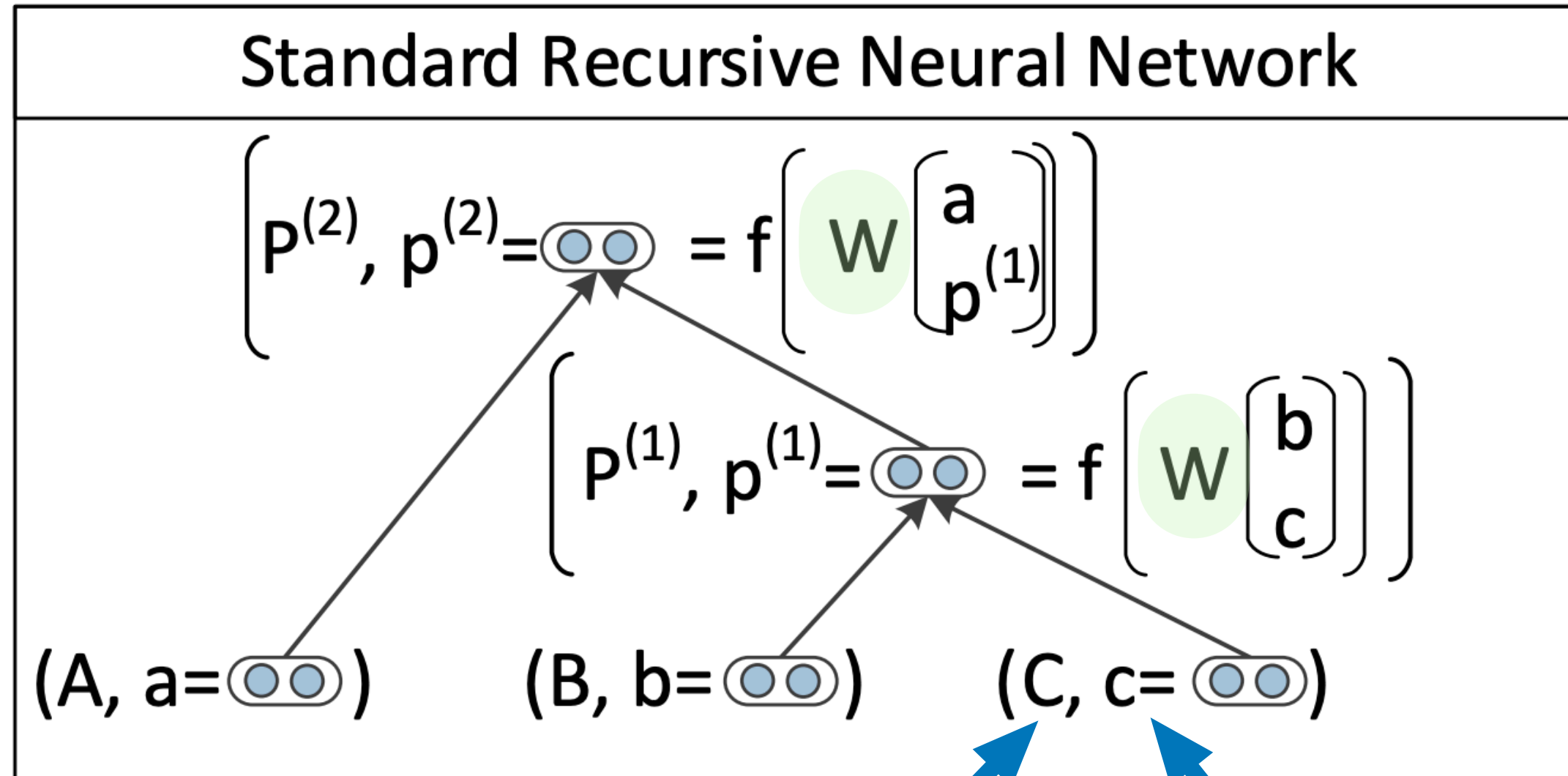
- Continuous representations for words and non-terminal nodes
- Compositional representations for non-terminal nodes
- Use neural networks to get compositional representations as well as scores for composition



Compositional Vector Grammar = PCFG + TreeRNN

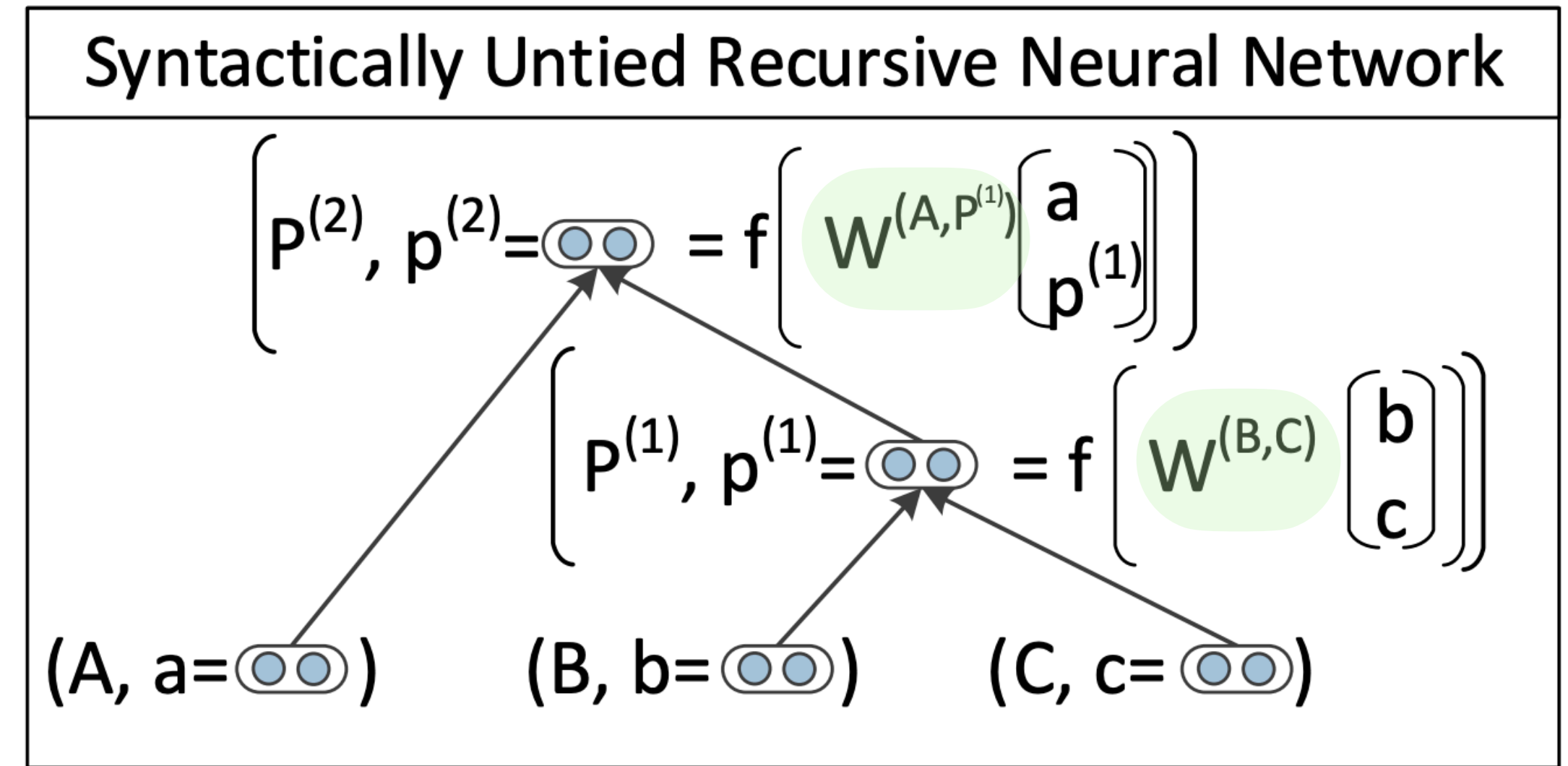
Recursive Neural Networks (Socher et al, 2013)

Weights depend on discrete
category of children (NP, VP)



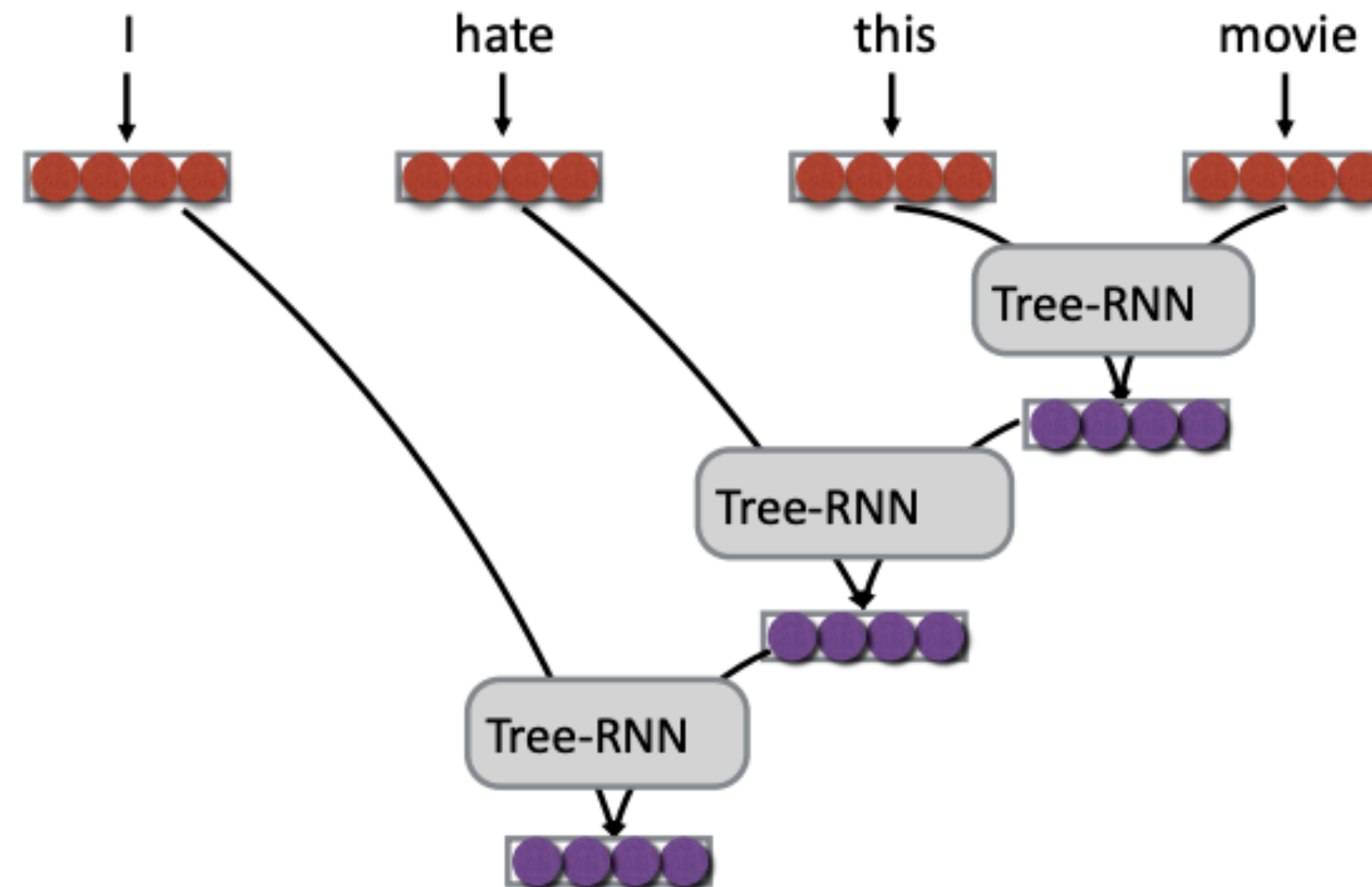
Node label Node embedding

Weights can be tied



or parameterized by constituency type

Recursive Neural Networks (Socher et al, 2013)



$$\text{tree-rnn}(\mathbf{h}_1, \mathbf{h}_2) = \tanh(W[\mathbf{h}_1; \mathbf{h}_2] + \mathbf{b})$$

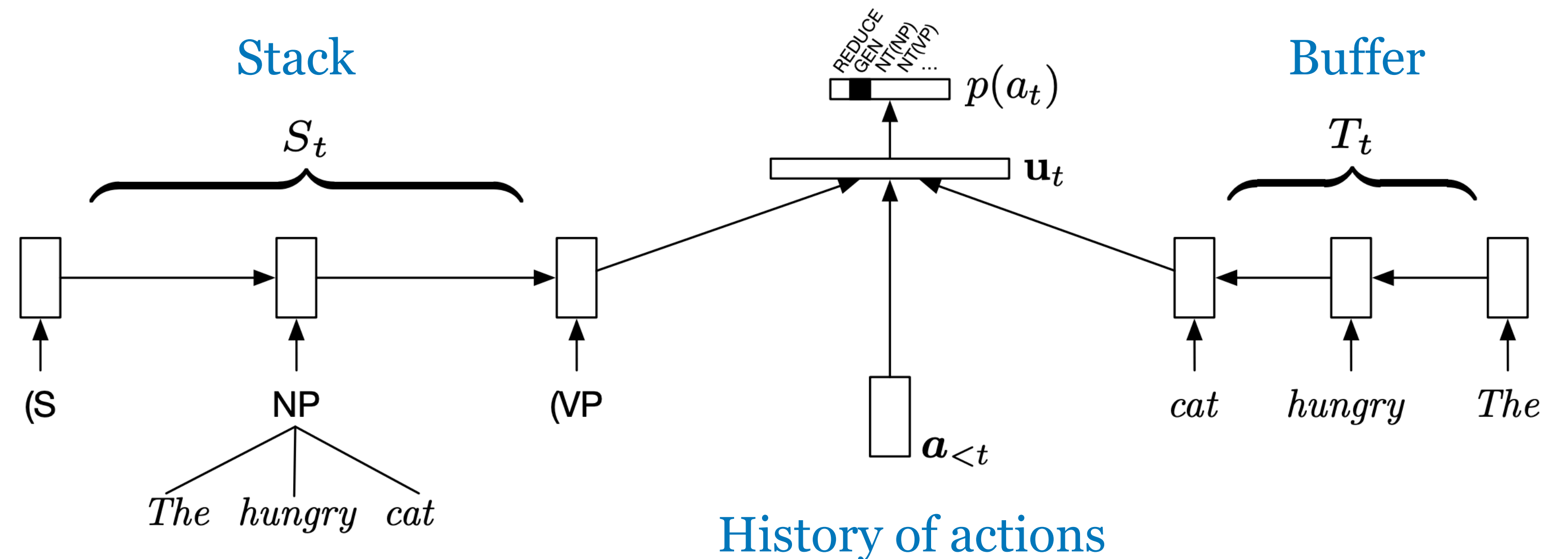
90.4 F1

Recurrent Neural Network Grammars (Dyer et al, 2016)

Transition Parsers

- Like Seq2Seq but output is a sequence of operations that builds the tree incrementally
- The sequence can guarantee structural consistency

Predict action from current configuration



Recurrent Neural Network Grammars

(Dyer et al, 2016)

Parser transitions

S: stack of open nonterminals and completed subtrees
B: buffer of unprocessed terminal symbols
x: terminal symbol
X: Non-terminal symbol
 τ : completed subtree

Before action			After action			
Stack _t	Buffer _t	Open NTs _t	Action	Stack _{t+1}	Buffer _{t+1}	Open NTs _{t+1}
S	B	n	NT(X)	S (X	B	n + 1
S	x B	n	SHIFT	S x	B	n
S (X τ_1 ... τ_ℓ	B	n	REDUCE	S (X τ_1 ... τ_ℓ)	B	n - 1

Top-down parsing

Input: <i>The hungry cat meows .</i>			
	Stack	Buffer	Action
0		<i>The hungry cat meows .</i>	NT(S)
1	(S	<i>The hungry cat meows .</i>	NT(NP)
2	(S (NP	<i>The hungry cat meows .</i>	SHIFT
3	(S (NP <i>The</i>	<i>hungry cat meows .</i>	SHIFT
4	(S (NP <i>The hungry</i>	<i>cat meows .</i>	SHIFT
5	(S (NP <i>The hungry cat</i>	<i>meows .</i>	REDUCE
6	(S (NP <i>The hungry cat</i>)	<i>meows .</i>	NT(VP)
7	(S (NP <i>The hungry cat</i>) (VP	<i>meows .</i>	SHIFT
8	(S (NP <i>The hungry cat</i>) (VP <i>meows</i>	<i>.</i>	REDUCE
9	(S (NP <i>The hungry cat</i>) (VP <i>meows</i>)	<i>.</i>	SHIFT
10	(S (NP <i>The hungry cat</i>) (VP <i>meows</i>) .		REDUCE
11	(S (NP <i>The hungry cat</i>) (VP <i>meows</i>) .)		

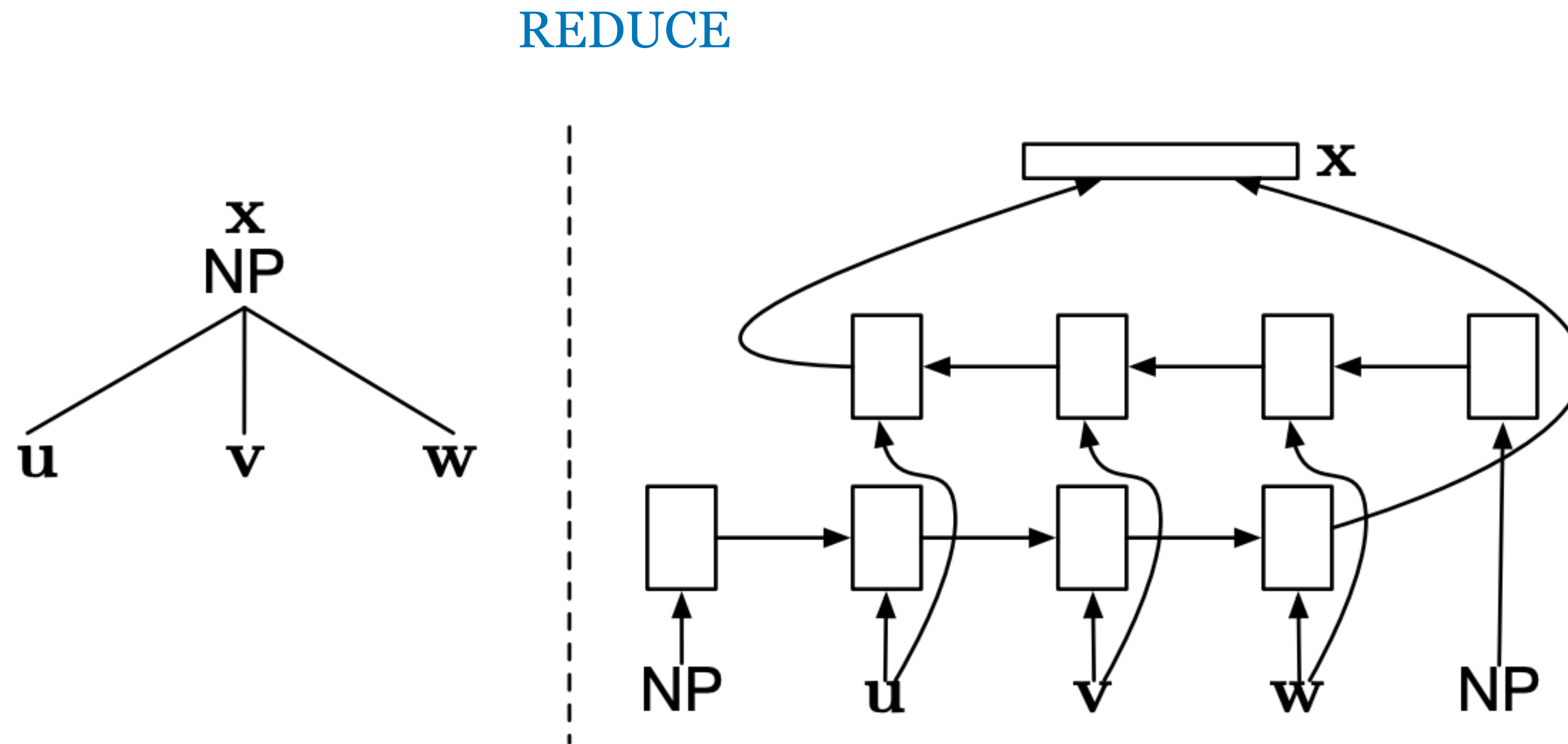
Actions:

NT(X): Open (create) a new non-terminal of type X

SHIFT: move x from buffer to stack

REDUCE: Close(finish) open non-terminal on stack

Recurrent Neural Network Grammars (Dyer et al, 2016)



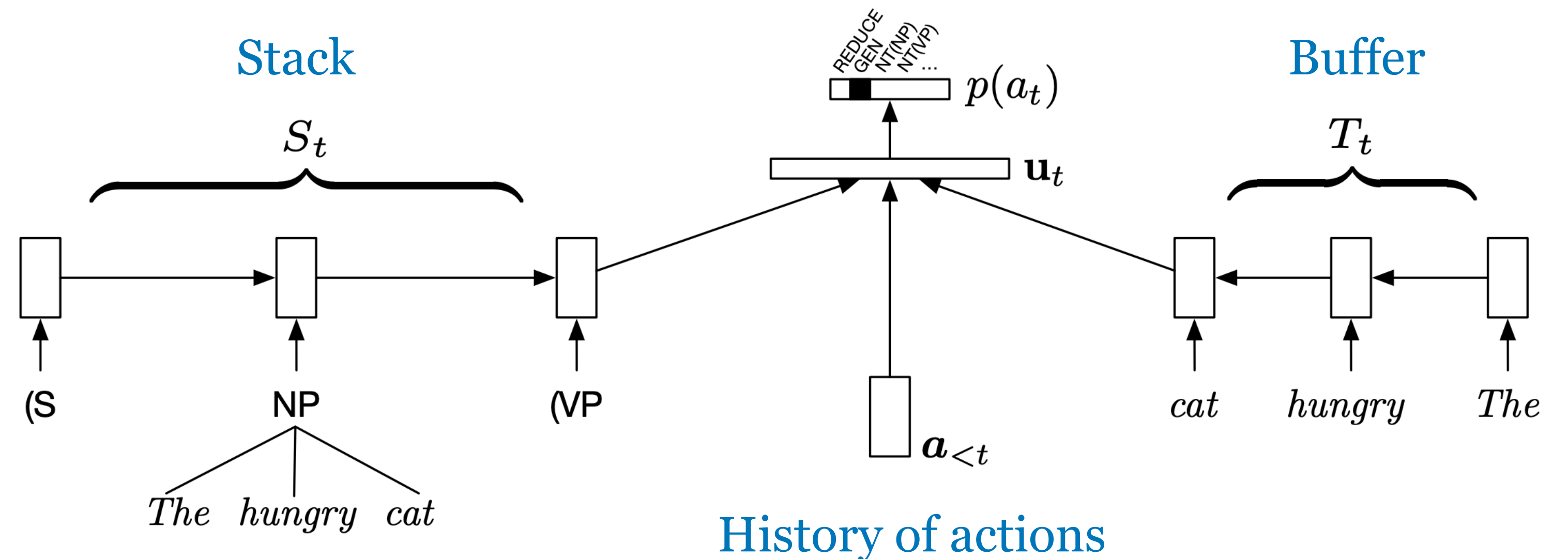
- BiLSTM to get composite representation of non-terminal

Recurrent Neural Network Grammars (Dyer et al, 2016)

Transition Parsers

- Like Seq2Seq but output is a sequence of operations that builds the tree incrementally
- The sequence can guarantee structural consistency

Predict action from current configuration



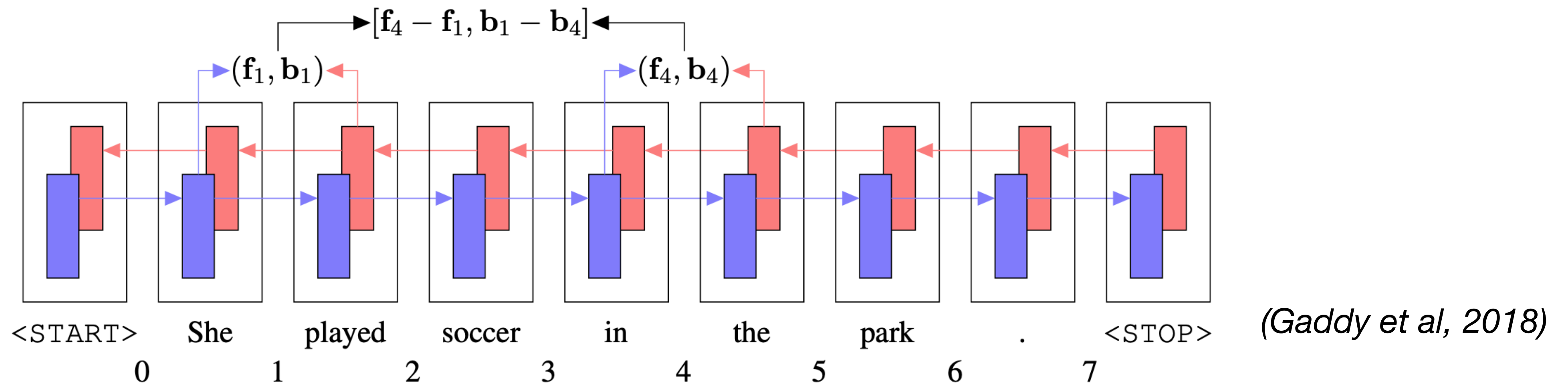
Span Labeling

(Stern et al. 2017)

- Simple idea: decide whether span is constituent in tree or not
- Scores labels and spans independently
- Allows for various loss functions (local vs structured), inference algorithms (CKY vs topdown)

- Word representation
- Span representation
- Label scoring

Span Labeling (Stern et al. 2017)



- Bidirectional LSTM to get forward/backward encodings (f_i, b_i) for position i
- Span (i, j) representation: concat vector differences $[f_j - f_i, b_i - b_j]$
- Feedforward neural networks to predict scores for labels and spans

$$S_{\text{labels}}(i, j) = \mathbf{V}_l g(\mathbf{W}_l \mathbf{s}_{ij} + b_l) \quad \text{vector} \quad S_{\text{label}}(i, j, l) = l \text{ th element of } S_{\text{labels}}$$

$$S_{\text{span}}(i, j) = \mathbf{v}_s^\top g(\mathbf{W}_s \mathbf{s}_{ij} + b_s) \quad \text{scalar}$$

Span Labeling (Stern et al. 2017)

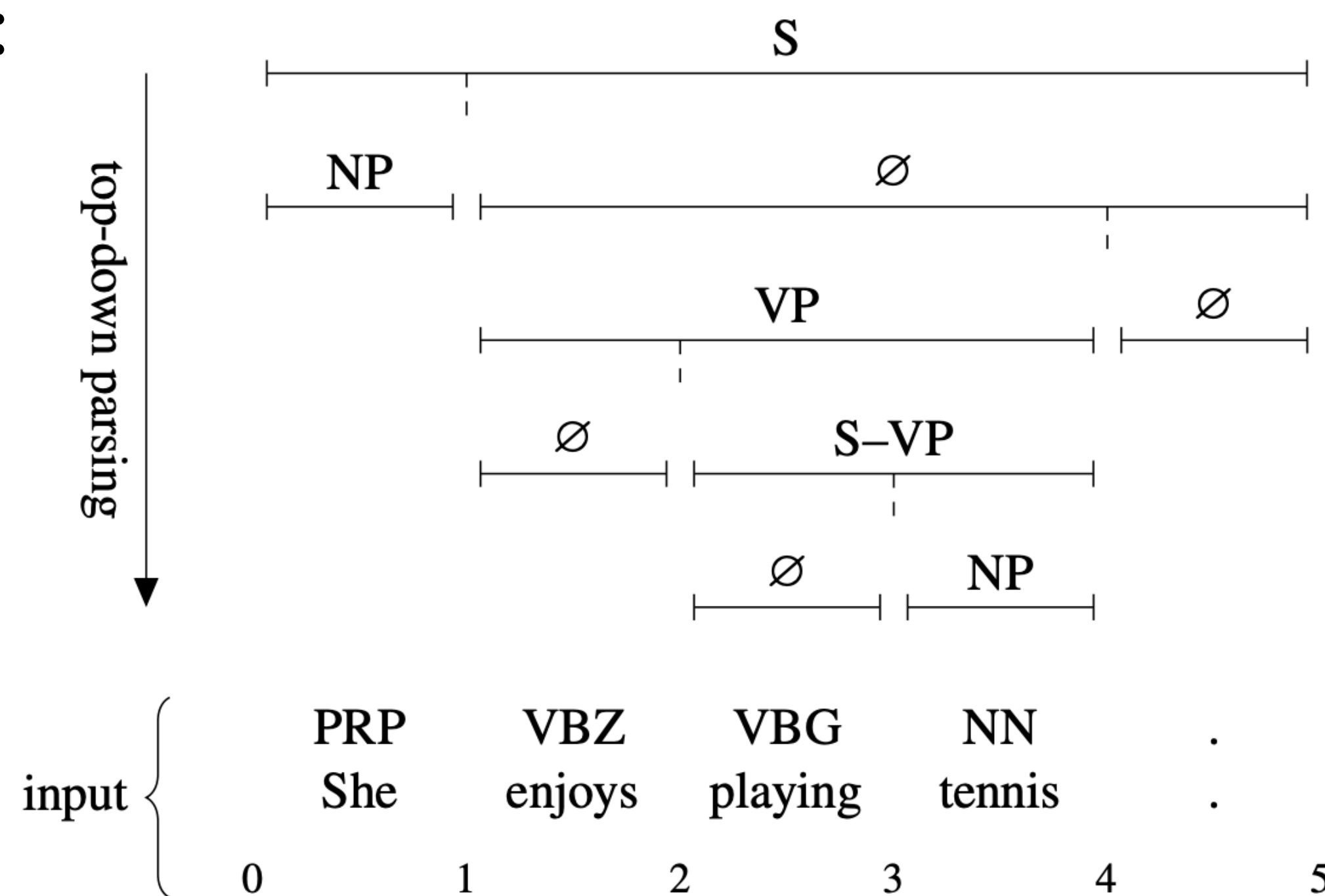
Greedy **top down parsing**

- Recursively for each span:
 - Assign a label
 - Pick a split point

$$\hat{l} = \arg \max_k S_{\text{label}}(i, j, l)$$

$$\hat{k} = \arg \max_k S_{\text{split}}(i, k, j)$$

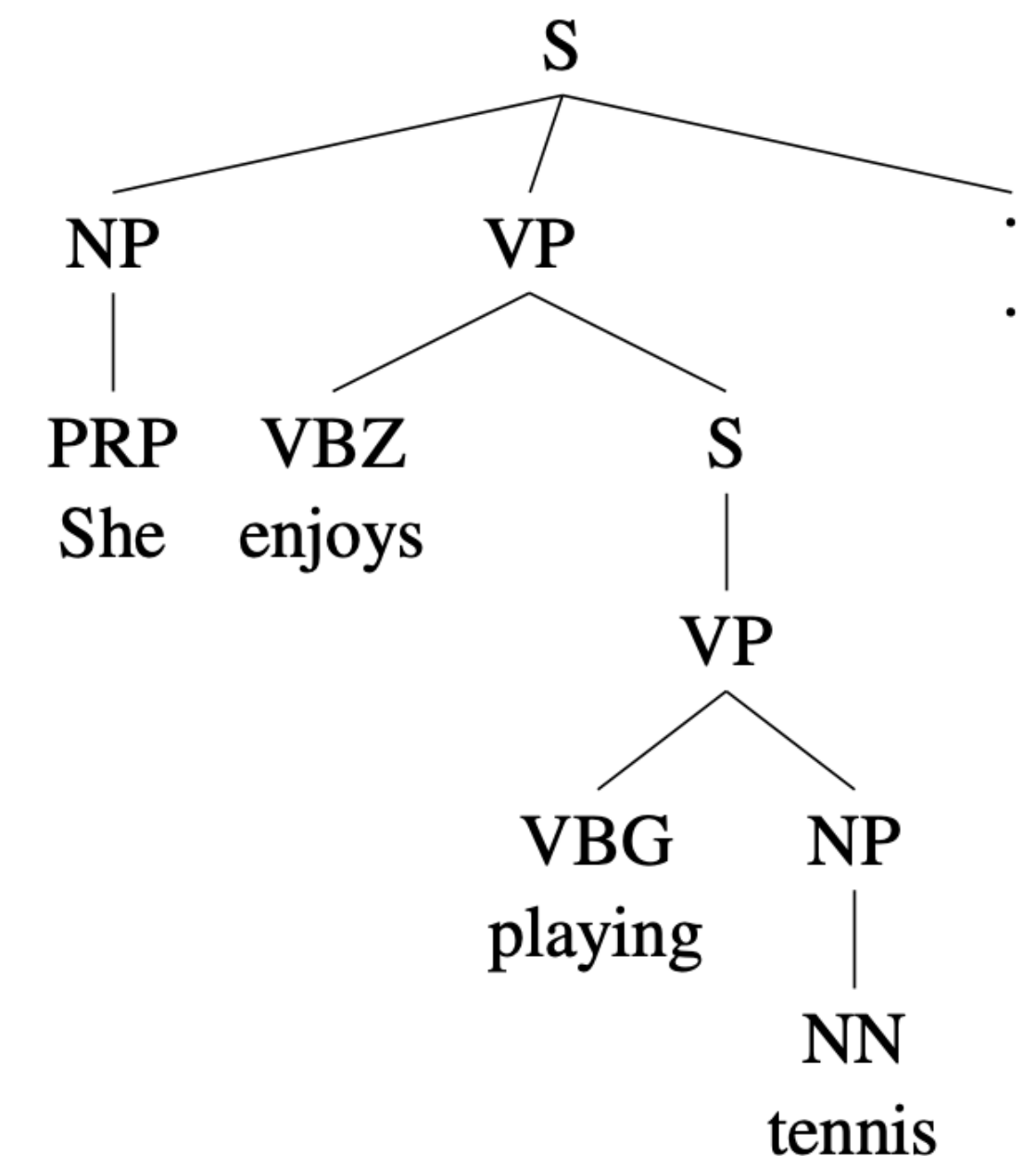
$$S_{\text{span}}(i, k) + S_{\text{span}}(k, j)$$



(a) Execution of the top-down parsing algorithm.

Running time?

$$O(n^2)$$

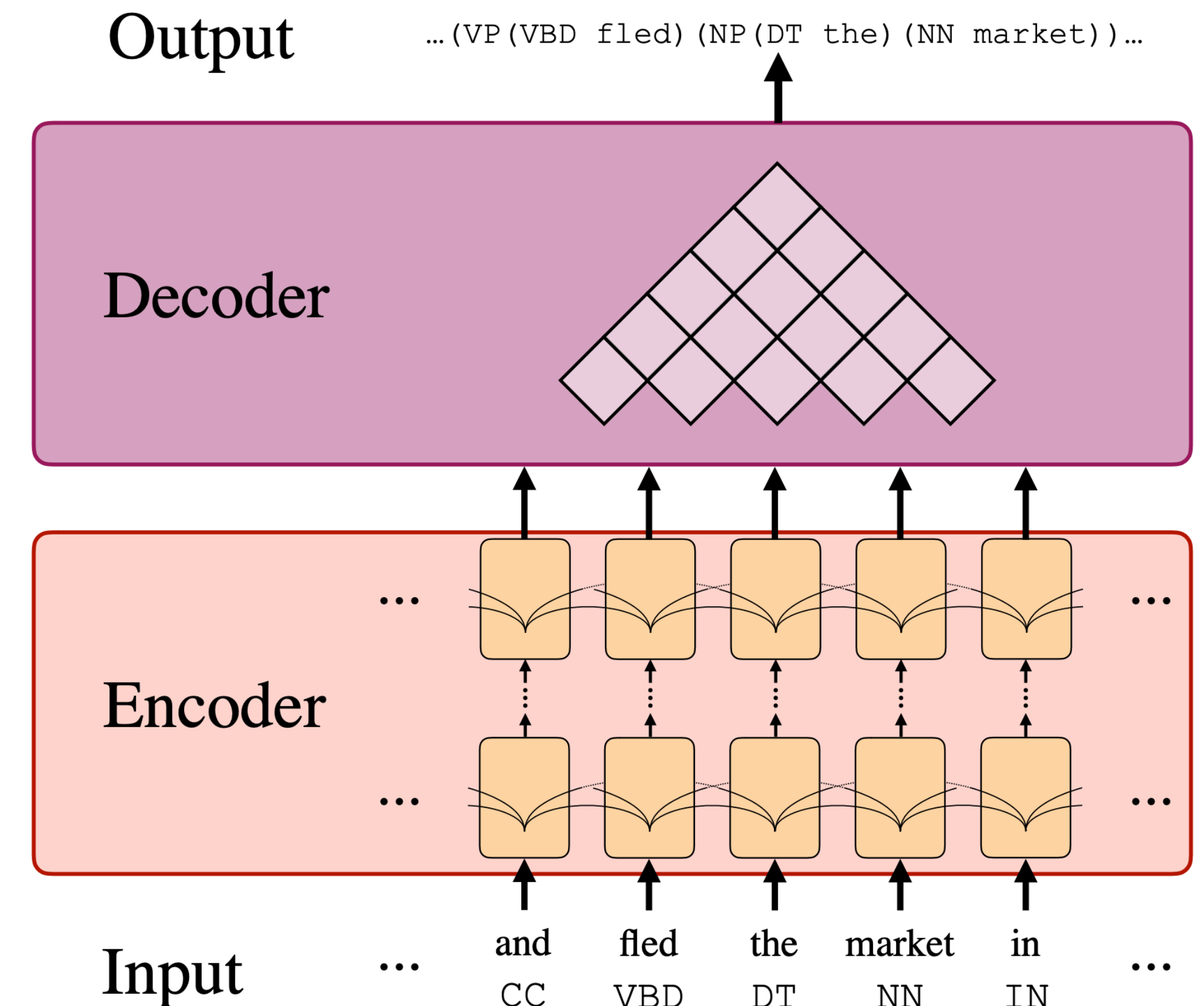


(b) Output parse tree.

91.8 F1

Self-Attentional Encoding (Kitaev and Klein, 2018)

- Self-attention based encoding
- Learned scoring $s(i, j, l)$ function for each span from token i to token j with label l
- CKY for decoding to find the best tree
- Berkeley neural parser: <https://github.com/nikitakit/self-attentive-parser>



93.6 F1

Self-Attentional Encoding (Kitaev and Klein, 2018)

- Improvements with pretrained representations

F1

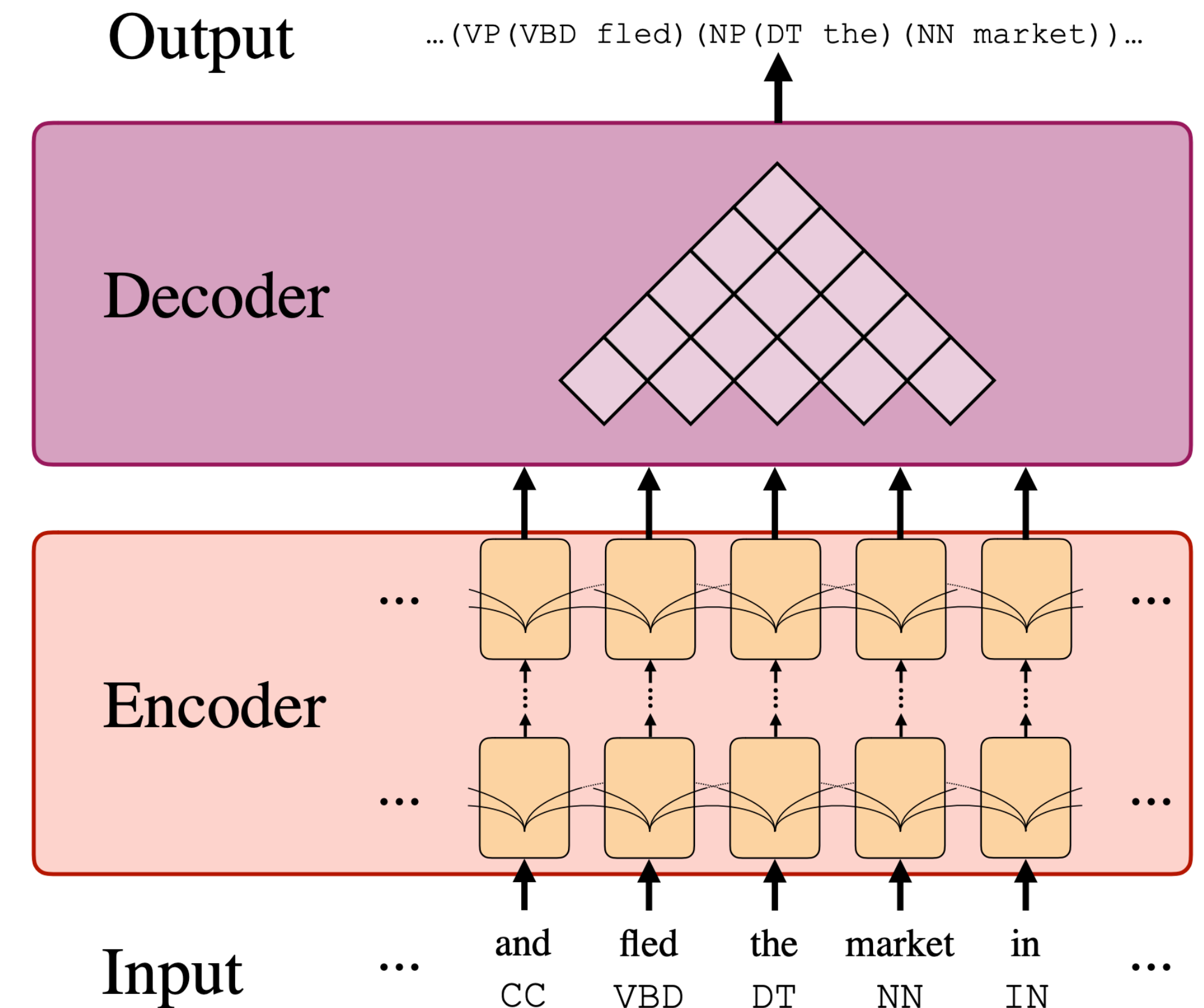
93.6 (no pretraining)

93.7 (w/ FastText)

95.2 (w/ ELMo),

95.7 (w/ BERT LARGE cased),

95.8 (Ensemble w/ BERT BASE/LARGE,
cased/uncased)



Summary

- Two types of structured representations: constituency vs dependency
- Formalism for context free grammars (CFG) and probabilistic context free grammars (PCFGs)
 - CFGs have terminals (leafs), non-terminals, and production rules
 - PCFGs are CFGs with probabilities on the rules
- Estimating probabilities for PCFGs and decoding (parsing)
- How to use neural networks for constituency parsing